

Passenger Queue Dynamics and Waiting-Time Simulation for Selected M66 Bus Stops



Project Documentation Submitted to the
Faculty of the School of Computing and Information Technologies

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Abstract

This simulation project examines whether hourly bus-trip allocation improves passenger waiting at five eastbound M66 stops. It combines recorded stop-hour demand with a frozen schedule in a discrete event model for separate morning and evening periods. Baseline, higher demand, reduced service, and combined conditions are tested. A genetic algorithm (GA) proposes integer allocations under fixed trip budgets and a mean final-backlog safeguard. Selected plans are frozen and compared with published service, a regularized reference, and a demand rule on matched unused passenger seeds. GA lowers mean completed wait against the regularized reference in all eight condition-period comparisons. Against the demand rule, two comparisons favor GA, two favor the rule, and four are inconclusive. Baseline morning shows a 0.113-minute allocation reduction within the 0.147-minute published-reference reduction. Its 0.034-minute timing contrast has an interval containing zero. Final backlog consists of arrivals at or after the last eligible bus in these tests. Local waits and period-end queues can worsen despite lower period means. The results support service comparisons under fixed assumptions; they do not establish optimizer superiority, robustness to changed inputs, or operational readiness.

Keywords: M66, passenger queues, discrete event simulation, genetic algorithm, service allocation, paired experiments

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1. Problem and Objectives

1.1 Project Context

This project focuses on passenger waiting and the use of available bus service across five consecutive eastbound M66 stops in Manhattan. Its real-world concern is whether service available at different times of day is aligned with changing passenger demand along the selected corridor.

The project team initially considered studying the EDSA Busway because it is locally relevant to commuters in the Philippines. The proposed queue simulation requires compatible stop-level boarding and alighting totals by hour together with a machine-readable bus schedule. During project scoping, the project team did not identify a public EDSA source that provided all these inputs at the required level of detail. The project selected M66 because the MTA publishes stop-hour ridership records and static schedules in the General Transit Feed Specification (GTFS) format (Metropolitan Transportation Authority [MTA], n.d.-a, n.d.-b). This was a data-feasibility decision and does not imply that the M66 corridor is more important than EDSA.

Passenger waiting is shaped by how quickly passengers arrive and how buses are spaced. Queues can grow between buses when demand rises or service is unevenly distributed across time, even when the total number of scheduled trips appears adequate. Timetables and aggregate ridership totals cannot directly reveal individual waits because those waits come from the interaction of passengers and service over time. Passenger-arrival research distinguishes the demand count from the timing process that places arrivals within an interval (Kieu & Cai, 2018). This distinction matters when hourly totals are used to simulate individual passengers.

The study covers one continuous eastbound segment, stop sequences 6 through 10, served by the same full-route M66 pattern. This keeps the physical stop order while making the study manageable. The weekday 06:00 to 11:00 and 15:00 to 20:00 periods are equal five-hour windows, allowing a consistent comparison of morning and afternoon-to-evening conditions. This controlled scope represents the selected segment and study periods rather than the full route or full service day.

1.2 Problem Statement

The available ridership and schedule records do not directly show individual waiting times, queues between buses, passengers still waiting at the end of service, or the onboard loads produced by service and demand together. This limits direct assessment of where demand and service are misaligned and whether redistributing existing trips can improve passenger outcomes.

1. **Unmeasured passenger queue performance.** Stop-hour counts and scheduled arrival times need to be connected through a simulation to estimate mean and 95th percentile (P95) completed waiting time, peak queue, period-end queue, final backlog, and bus occupancy at the five selected stops. These are modeled outcomes, not waiting-time observations contained in the source records.
2. **Vulnerability to operating stresses.** The records alone do not establish how a 40 percent increase in selected-stop demand, removal of one trip from each study hour, or their combination changes waiting, queue growth, backlog, and occupancy relative to baseline service. Consistent scenario rules and matched replications are needed to isolate those changes.
3. **Risk of inefficient hourly service allocation.** A fixed number of trips can be distributed across hours in ways that produce different waits. A feasible allocation method is needed to test whether the

same scenario-specific trip budget can reduce completed waiting time while controlling final backlog. Light occupancy and long waits can coexist, so higher occupancy alone is not evidence of a better passenger outcome.

1.3 Objectives

The project objective is to implement and evaluate a reproducible AI-supported discrete event simulation for five consecutive eastbound M66 stops under normal demand, 40 percent higher selected-stop demand, one fewer trip in each study hour, and the combination of higher demand and reduced service. This simulation project evaluates feasible hourly trip allocations within each condition-specific fixed service budget. The completed implementation and results in this paper provide the evidence for these objectives.

1. Implement and check separate morning and evening simulations for the five selected stops. Estimate mean and P95 completed waiting time, peak queue, period-end queue, final backlog, and bus occupancy using at least 30 independent passenger-arrival replications. Report 95 percent confidence intervals for the stated scenario and service comparisons, and distinguish implementation checks from validation against actual queues.
2. Quantify changes from the normal-demand reference caused by 40 percent higher selected-stop boarding demand, one fewer trip in every study hour, and the combined condition. The service reduction changes the morning budget from 43 to 38 trips and the evening budget from 44 to 39. Use consistent passenger-generation, load, boarding, and boundary rules to compare waiting, queues, backlog, and occupancy.
3. Use the selected genetic algorithm to produce feasible integer trip allocations across the five study hours for baseline and all three stress conditions. Freeze each selected plan and compare it with its matching published reference, regularized reference, and demand rule on paired unused-seed replications with 95 percent confidence intervals. Assess whether completed waits decrease without a supported increase in final backlog, report the trip counts and candidate headway rule, and examine queue and occupancy changes and stop-level tradeoffs.

1.4 Significance of the Simulation

The simulation contributes a reproducible way to connect recorded passenger activity and scheduled service to modeled waiting outcomes. Its practical value is the ability to test a service change before a field trial, keep the resource budget explicit, and reveal cases where a better period average hides a worse outcome at one stop. It produces conditional planning evidence; it does not establish that a selected allocation is ready to operate.

Transport Planners and Service Analysts. Use the method to compare fixed-budget allocations and simple alternatives under the same passenger streams. For example, the reduced-service morning results identify a larger waiting penalty and a larger potential benefit from changing the schedule than the baseline results. The control comparisons help separate benefits from regular spacing and benefits from hourly reallocation. Operational use would require the further evidence described in the recommendations.

Passengers. They are the intended indirect beneficiaries if a tested allocation later passes field and operational checks. Shorter and more reliable waits could make public transport easier to use. The current model estimates completed waits and unserved backlog, so it cannot claim measured accessibility gains or

benefits for a particular passenger group. Local increases at some stops make it necessary to examine who may lose service quality when an average improves.

Students and Researchers. Use the shared event engine, frozen inputs, saved plans, and paired comparisons as a transparent example of simulation-based optimization. The project also shows that an AI method should be checked against a simple rule: GA is better in some demand-rule comparisons and worse in others. This makes the contribution a testable method and an evidence set rather than a claim that AI always wins.

Sustainable Development Goal 11. The study relates to Target 11.2 on safe, affordable, accessible, and sustainable transport systems (United Nations, n.d.). Waiting and service availability are relevant to practical use of public transport. This simulation project does not measure progress toward the target, emissions, affordability, ridership growth, or population-level accessibility.

2. Scope and Limitations

2.1 Scope

The simulation models weekday passenger activity at eastbound M66 stop sequences 6 to 10 in Manhattan, with separate five-hour morning and evening experiments. Historical demand comes from the cleaned 2025 extract. Service comes from one frozen 2026 GTFS pattern. Four demand-service conditions and four service methods are evaluated. The decision variable is five hourly integer trip counts, constrained by the period budget and hourly bounds.

The scope is a controlled service-allocation study. Upstream Stops 1 to 5 provide aggregate initial-load estimates, while their queues are outside the modeled segment. Passenger arrivals within each hour are generated from fixed counts. Queues start empty, and eligible trips can complete downstream events after the official period boundary. These boundaries make scenario comparisons consistent and keep the conclusions specific to the experiment.

2.2 Limitations

The data contain boardings and alightings, not measured queue-entry times or actual passenger waits. The historical and schedule inputs come from different years. Available-record means, fixed rounded demand counts, uniform within-hour timing, assumed capacity, and aggregate alighting targets are explicit approximations. These limits prevent interpreting the output as a reconstruction of an observed operating day.

The model excludes random travel delays, dwell-time feedback, bus bunching, full-route vehicle circulation, driver duties, passenger destinations, and accessibility needs. A fixed trip budget does not establish equal operating cost or fleet feasibility. Confidence intervals cover generated passenger-timing variation under fixed inputs, not demand-estimation or model uncertainty. The limited search and small technique screen establish neither optimizer superiority nor a global optimum. Robustness to changed capacity or arrival patterns has not been tested. Section 4 states the model rules and their basis; Section 9 describes the further evidence needed.

3. Dataset Analysis

The study boundaries define which records are needed. This section explains how the two public sources are prepared, checked, and connected to the model. Demand records supply passenger inputs, while GTFS supplies service times and route order.

3.1 Dataset Sources

The demand source is MTA Bus Stop Level Ridership: Beginning 2024, published through New York State Open Data. A source record contains route, direction, stop, date, hour, boardings, alightings, and trips (MTA, n.d.-b). The frozen CSV contains the 2025 M66 eastbound extract. Boardings and alightings are recorded passenger activity, not individual waiting-time measurements. Historical Trips describe the source data and support quality checks; they do not determine simulated bus arrivals.

The service source is the frozen Manhattan bus GTFS feed from MTA developer resources (MTA, n.d.-a). Feed version `gtfs_m_20260611T155328Z` covers June 27 to September 5, 2026. The study selects M66 eastbound service ID `MQ_C6-Weekday`, activated on 14 dates in `calendar_dates.txt`, including June 29. Calendar and exception records define its operating dates (MobilityData, n.d.). This pattern supplies a consistent timetable for full-route trips serving all five verified stops; it defines one controlled reference rather than a typical weekday or a reconstruction of 2025 service.

Table 1. Verified cross-source stop mapping

Seq.	2025 ridership stop ID	Frozen GTFS stop ID	Verified stop name
6	403492	403492	W 65 St / Central Park West
7	403494	403494	E 65 St / 5 Ave
8	404994	400027	Madison Ave / E 65 St
9	403495	405489	Madison Ave / E 67 St
10	403497: Jan 1 to Mar 28; 405627: Mar 31 to Dec 31	405627	E 68 St / Lexington Ave

Note. Sources: frozen 2025 ridership CSV and `gtfs_selected_stop_mapping.csv`. Dates describe the observed records for Stop 10. Route, direction, stop sequence, and checked names support the mapping; an unchecked stop ID alone would miss changed identifiers.

3.2 Key Variables

Table 2. Ridership variables and their roles

Variable	Meaning and use
Date and Hour	Define year, weekdays, study periods, and stop-hour averages.
Route ID and Direction	Filter M66 and eastbound records.
Stop Sequence and Stop Name	Select the segment and verify route order and cross-source mapping.
Stop ID	Identify source stops; IDs are checked because they can change between sources or dates.
Boardings	Provide the stop-hour demand proxy used to generate passengers.
Alightings	Provide hourly alighting targets and support upstream net-load estimates.
Trips	Describe historical source coverage and identify impossible zero-trip activity.

Table 3. GTFS variables and their roles

Variable	Meaning and Project Use
feed_version	Identifies the frozen schedule snapshot, gtfs_m_20260611T155328Z.
route_id and direction_id	Select route M66 and direction 0, verified as eastbound through the pattern and stops.
service_id and calendar_dates	Select MQ_C6-Weekday and check its 14 activation dates rather than assuming the name establishes daily service.
trip_id	Identifies one bus across its stop events and keeps trip removal consistent at all five stops.
trip_headsign	Checks the eastbound destination, EAST SIDE YORK AV CROSSTOWN.
shape_id	Selects full-route pattern M660030 and excludes short-turn M660028, which does not serve the full selected corridor.
stop_id, stop_name, and stop_sequence	Verify physical stops and route order. Sequences 6 to 10 define the selected segment; verified upstream stops support initial loads.
arrival_time	Creates the published reference events, hourly trip counts, and downstream travel offsets used by candidate schedules.
departure_time	Supports schedule inspection. The event engine uses arrival_time and does not model a separate random dwell-time process.

Note. Sources: frozen feed_info.txt, trips.txt, stop_times.txt, calendar_dates.txt, and stops.txt; GTFS schedule reference (MobilityData, n.d.). These variables describe scheduled service, not observed bus operations.

3.3 Dataset Quality Issues and Treatment

Automatic Passenger Counters (APC) provide the boarding and alighting estimates described in the source documentation. These counts record passenger activity on buses, rather than individual queue-entry times.

The data preparation process filters route, direction, year, weekdays, study hours, and selected stop sequences; parses date and hour fields; checks missing cells, duplicates, negative counts, and inconsistent passenger activity; and verifies GTFS stop mappings. Appendix D gives the filtering counts and source-acquisition screenshots. The cleaning rule excludes one record with zero Trips and two alightings. The analysis retains valid large boarding counts because size alone does not make a record incorrect.

There are no missing cell values, exact duplicates, logical duplicates, or negative counts in the selected records. A complete date-stop-hour grid has 13,050 expected combinations. The source is missing 235 combinations, and excluding the invalid record leaves 236 unavailable combinations after cleaning. The analysis computes means from available valid observations. The analysis does not treat missing combinations as zero boardings. Missing coverage can still bias a mean if the unavailable records differ from those observed.

Table 4. Dataset quality issues and treatment

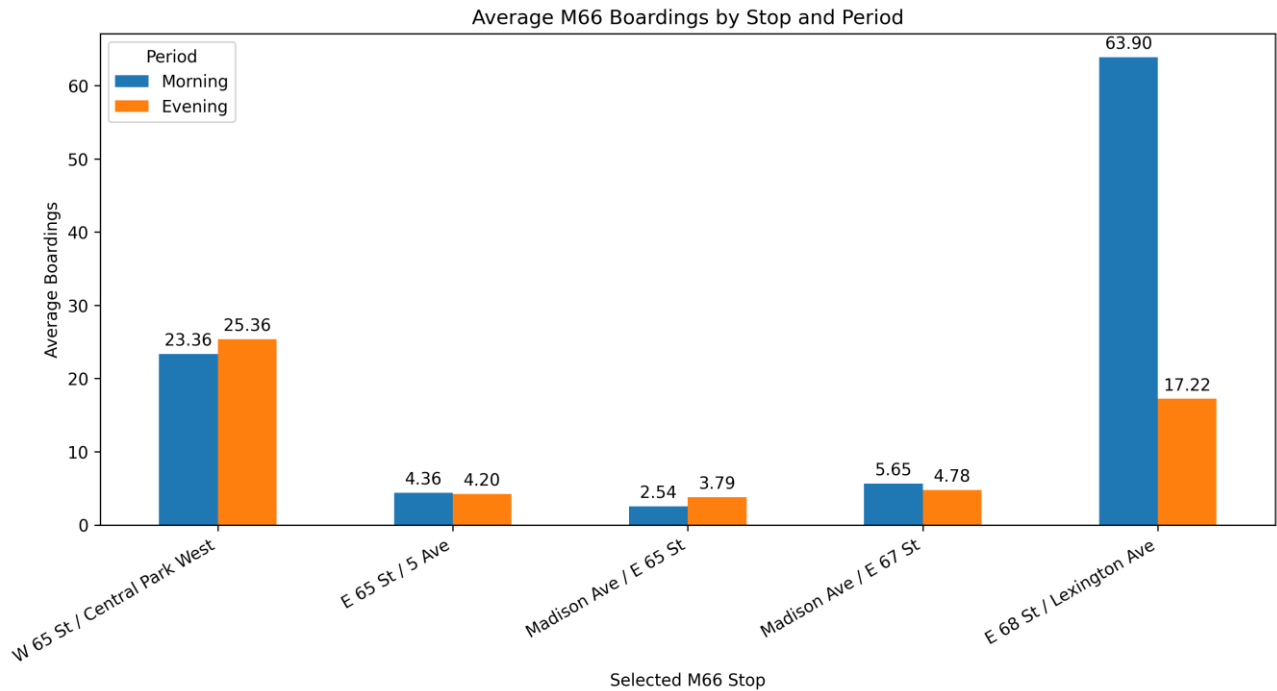
Issue	Extent	Treatment and Model Effect
Passenger activity with zero Trips	1 selected-stop record; 3 records at the verified physical upstream stops.	Exclude records with positive Boardings or Alightings and zero Trips because these fields conflict. The selected record has two alightings. Apply the same rule before estimating upstream flow.
Source-absent combinations	235 absent before cleaning; 236 unavailable after cleaning, out of 13,050 expected date-stop-hour combinations.	Use available valid observations to calculate means. Do not fill absent combinations with zero, since absence can reflect APC coverage or zero APC counts. Uneven missingness can bias representative demand.
Incomplete APC coverage	A source-level limitation, not a missing cell in a saved row.	Treat observed boardings as a demand proxy, not a full count of everyone who wanted to board. Unrecorded activity and passengers who never boarded are not recovered by cleaning.
Changed stop identifiers	Historical and frozen GTFS identifiers differ at sequences 8 to 10; Stop 10 also changes within 2025.	Verify route, direction, sequence, and stop name together using Table 1. This connects the physical corridor without dropping historical records because an identifier changed.
High but plausible counts	Maximum selected-stop Boardings is 206.	Keep large counts when no consistency check identifies an error. Automatic trimming could remove genuine busy hours and lower the demand input.
Demand and schedule reference periods	2025 ridership and the frozen 2026 GTFS pattern.	Interpret outcomes as controlled scenario comparisons. The inputs do not reconstruct an observed operating day, and the selected service ID is not valid on every weekday.
Missing cells, duplicates, and negative counts	Zero missing cells, exact duplicates, Date-Hour-Stop Sequence duplicates, and negative counts in selected records.	No correction is needed for these checks. Passing them establishes internal consistency, not complete coverage or field validity.

Note. Sources: frozen ridership CSV, data_quality_summary.csv, the dataset and baseline notebooks, and MTA dataset documentation (MTA, n.d.-b). Upstream checks use the five verified physical stop IDs from the full-route pattern.

The 235 source-absent combinations occur across 74 weekdays. Stop 7 has the largest number, 79, and the hour starting at 19:00 has 77. The absence pattern is uneven, so missing-cell checks alone would give an incomplete picture of dataset quality. The dataset analysis keeps the distinction between an absent row, an invalid row removed by a rule, and an observed zero.

3.4 Demand Patterns

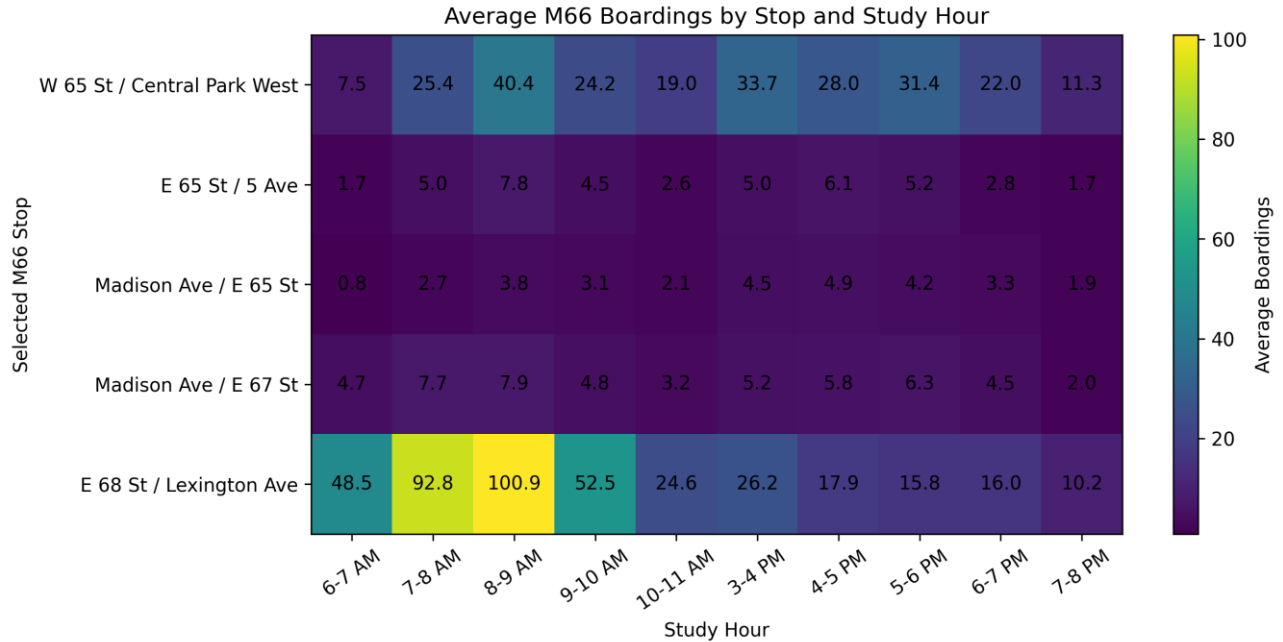
Figure 1. Mean recorded boardings by stop and period



Note. Source: stop_period_summary.csv. Each value is a mean per observed stop-hour, not a five-hour total. Morning and evening means use their available records.

The first observation is spatial concentration. Stop 10 has the largest morning mean, 63.90 boardings per observed stop-hour, while Stop 6 has 23.36. Stops 7 to 9 have much smaller morning boarding means, from 2.54 to 5.65. This means a network average alone would hide the main concentration of boarding demand.

Figure 2. Mean recorded boardings by stop and study hour



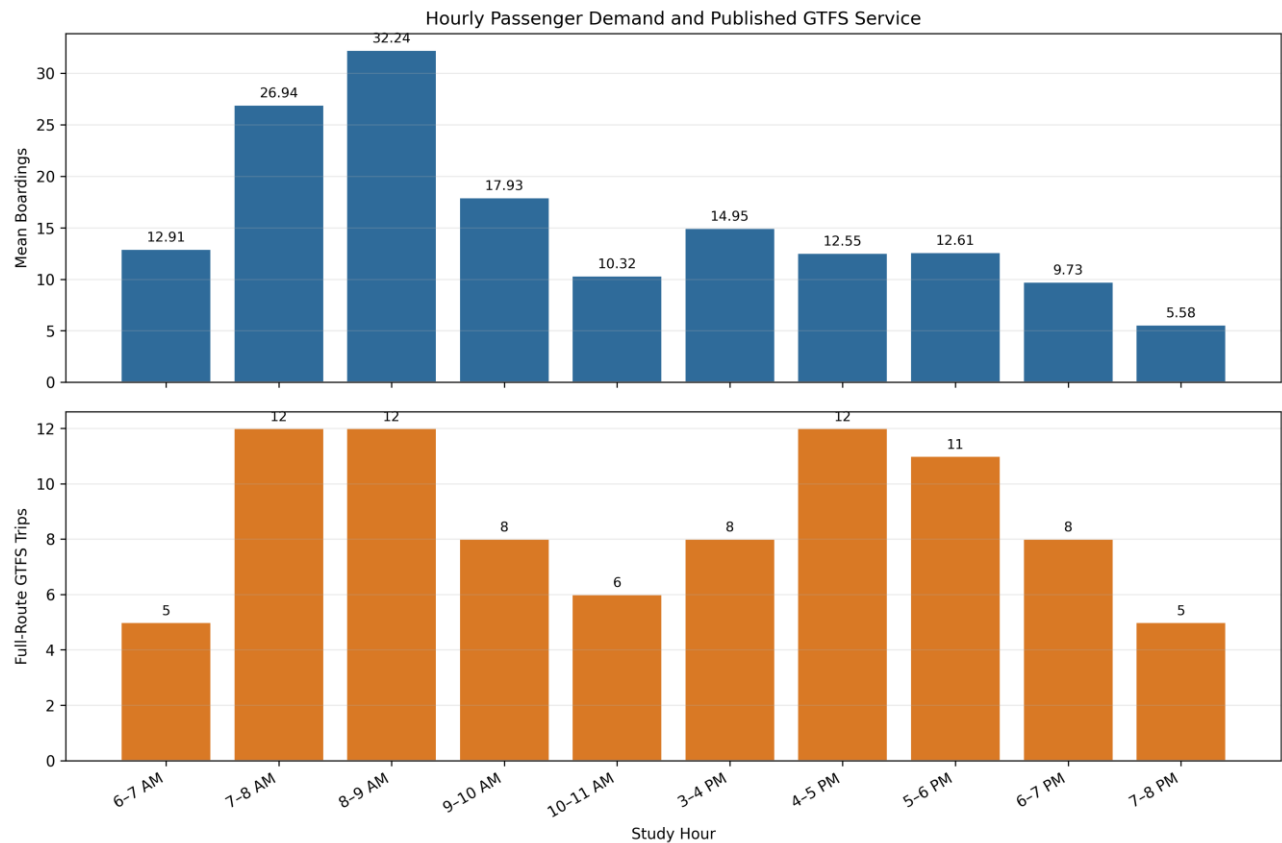
Note. Source: cleaned selected ridership CSV, grouped by stop sequence and hour. Cells show repository chart values rounded to one decimal place; analysis uses unrounded means. The plot describes historical boardings and does not show measured queues.

The second observation is temporal variation. The mean across observed selected-stop records is highest in the hour starting at 08:00, at 32.24 boardings, and falls to 10.32 at 10:00. Evening values fall from 14.95 at 15:00 to 5.58 at 19:00. Figure 2 shows that the stop and hour patterns interact, especially at Stop 10 in the morning.

The third observation is the change in boarding and alighting roles. Stop 10 has an evening mean of 17.22 boardings and 41.72 alightings per observed stop-hour. Stop 6 has 25.36 evening boardings and 2.38 alightings. A stop with many alightings can free space for passengers further along the segment. The model uses both flows rather than assigning every bus an empty starting load.

These patterns support using separate stop-hour demand cells and hourly service decisions. They do not prove that a proportional demand allocation is optimal. Passenger timing, downstream travel offsets, and the period boundary also affect waiting. The evaluation tests a proportional demand rule as an explicit control.

Figure 3. Hourly demand and published service counts



Note. Source: repository output/figures/06_hourly_demand_and_gtfs_service.png. The upper panel averages boardings across observed selected-stop records; the lower panel counts eligible trips at Stop 6. Separate units avoid interpreting demand and trips as the same quantity.

The first morning hour has five trips, while the next two have twelve each. Demand also changes across these hours. This pattern motivates testing alternative allocations; the records do not establish why the operator selected those counts.

3.5 Data Support for the Model

The cleaned ridership records provide separate boarding and alighting means for each selected stop and hour. Boardings become rounded passenger counts, while alightings become per-hour targets. A verified upstream subset for the physical Stops 1 to 5 provides the aggregate net flow used to initialize buses before Stop 6. These separate inputs preserve the different boarding and alighting roles visible in the data.

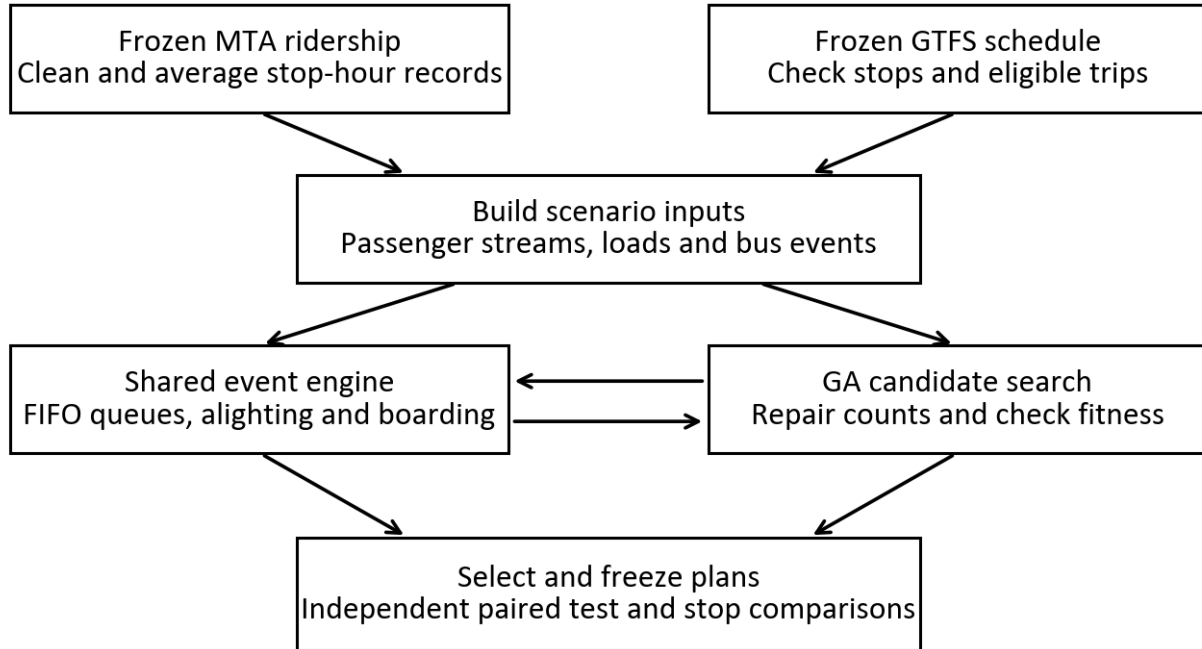
GTFS provides the ordered stop mapping, eligible full-route trips, published reference event times, hourly service budgets, and median downstream time offsets. A historical Trips value in the ridership CSV is not substituted for a GTFS event. Keeping these roles separate prevents historical source coverage from becoming a simulated timetable.

The two sources support the service-allocation experiment: ridership supplies passenger inputs, while GTFS supplies scheduled service.

4. System and Simulation Design

The demand patterns raise a question about service timing and allocation. The design translates that question into a terminating discrete event model, defines what changes during a run, and makes the assumptions explicit before describing the event loop.

Figure 4. Data preparation simulation and service optimization workflow



Note. The GA proposes hourly counts, receives fitness from the shared event engine, and repeats within its search budget. The final test uses frozen plans and unused passenger seeds. Reference and control schedules use the same engine.

4.1 Simulation Question

For the five selected eastbound M66 stops during weekday 06:00 to 11:00 and 15:00 to 20:00 periods, how do a 40 percent increase in selected-stop demand, removal of one trip from every study hour, and their combination change mean and P95 completed waiting time, queues, final backlog, and occupancy relative to normal-demand service? Can a genetic algorithm redistribute the same condition-specific number of trips across the five hours to reduce completed waiting time without worsening mean final backlog relative to the matching reference, and how does it compare with regularized service and a simple demand rule?

4.2 System Representation

The modeled system has passengers, buses, and five stop queues. A passenger belongs to one stop and joins its queue at a generated arrival time. A bus has a trip ID and an onboard count. It visits the stops in route order. At a bus event, eligible onboard passengers alight before waiting passengers board. Passengers board in first-in, first-out (FIFO) order until capacity is reached. People who cannot board remain in the queue.

The morning and evening simulations are separate terminating experiments, each starting with empty selected-stop queues. There is no warm-up period and no queue carried from morning to evening. This

initialization defines a study-window comparison; it may miss people already waiting when a real study period begins.

4.3 State Variables

Table 5. Simulation states inputs and recorded values

Item	Role	Type and Initial Value
simulation_clock	Time of the current event in minutes from midnight.	Float; 360 for morning and 900 for evening.
simulation_events	Pending bus and passenger events ordered by time and priority.	Heap built from the passenger stream and schedule.
stop_queues	FIFO passengers and their arrival times at each stop.	One Python list per selected stop; empty at period start.
Queue length	Number currently waiting at a stop.	Derived integer; zero at period start.
bus_occupancy	Passengers onboard each modeled trip.	Integer; begins with the assigned upstream load.
Available space	Capacity minus occupancy after alighting.	Derived nonnegative integer; updated at each bus event.
Assigned and actual alightings	Assigned target and the count that can actually alight.	Target is a fixed input; actual count is capped by current occupancy.
Demand counts and capacity	Inputs held fixed within the replication.	Rounded stop-hour counts and capacity of 60.
Waiting records	Stores arrival time, boarding time, and completed wait for each passenger who boards.	Records added when a passenger boards.
Queue and bus histories	Stores queue and occupancy values after processed events.	Records added as events are processed.
Peak queue	Tracks the largest observed stop queue during a run.	Derived from event history; starts at zero.
Period-end queue	Stores the queue snapshot at the official boundary.	Recorded at 11:00 or 20:00.
Final backlog	Records the total number of passengers still waiting after eligible events finish.	Recorded after all eligible bus events finish.

Note. Source: m66_core.py. Inputs and recorded statistics are distinguished from changing simulation states.

4.4 Modeling Assumptions and Basis

The assumptions below connect aggregate passenger activity and scheduled service to individual simulation events. Each entry states the rule, its basis, and its effect on interpretation.

Table 6. Modeling assumptions basis and possible effects

Assumption	Basis	Possible Effect
Boarding limit of 60 passengers per bus	Trip-specific vehicle capacities are absent from the inputs. The model uses an explicit experimental limit rather than infer a physical capacity from ridership.	Capacity-limited boarding may differ when actual vehicles have another capacity.
Observed boardings represent demand	The source records boardings rather than queue-entry times or everyone who wanted to board. Boardings provide a reproducible stop-hour proxy (MTA, n.d.-b).	Demand may be understated when people cannot board or are not observed.
Fixed rounded stop-hour passenger counts	The model uses floor(mean + 0.5). High demand multiplies the unrounded mean by 1.40 before rounding.	Counts vary by stop and hour but do not vary between replications. Low counts can have uneven proportional changes.
Uniform passenger timestamps within each hour	No individual arrival timestamps are supplied. Uniform times implement the hourly counts; finer stochastic models need finer evidence (Kieu & Cai, 2018).	Bursts and schedule-aware arrival behavior are omitted.
Empty queues at the start of each period	The source does not measure people waiting just before 06:00 or 15:00. Separate terminating runs use the same initial rule.	First-hour waiting may be understated if an initial queue exists.
FIFO boarding with no abandonment	No priority, abandonment, or route-switching records are available. FIFO makes the boarding rule explicit.	Real passengers may leave, change routes, or have different boarding priority.
Reference buses use published GTFS times	The frozen schedule supplies timetable events; actual bus arrival observations are outside the inputs (MTA, n.d.-a).	Traffic delays, early running, and bunching are absent.
One eligible trip ID carries one bus through the segment	Full-route pattern M660030 serves all five stops; its trip ID connects their events. Short-turn M660028 is excluded.	Full-route circulation is outside scope, and service specific to Stop 10 may be understated.
Aggregate upstream load distributed among buses	Net mean flow at upstream Stops 1 to 5 is clipped nonnegative, rounded once per hour, and divided by quotient and remainder with trip ID tie ordering.	Starting loads are approximations; hourly totals are preserved when trip counts change.
Aggregate alighting targets distributed by entry hour	Each target is assigned across eligible buses by quotient and remainder. The Stop 6 entry hour is used at every stop; actual alightings cannot exceed occupancy.	Trip destinations are omitted, and downstream clock-hour flows can differ from target assignment.
Candidate buses use midpoint spacing	Hourly counts require exact event times. Equal midpoint spacing makes candidate construction deterministic and comparable across methods.	GA versus published service includes both spacing and hourly allocation effects.

Assumption	Basis	Possible Effect
Candidate travel offsets use hour-specific GTFS medians	Median offsets preserve typical scheduled progression from Stop 6 for the entry hour.	Trip-specific running time and stochastic travel or dwell delays are omitted.
High demand changes selected-stop boardings only	The 1.40 multiplier is a controlled pressure test. Upstream and alighting hourly targets remain fixed.	The stress test does not represent a route-wide increase in every passenger flow.
Reduced service removes the trip nearest the half-hour	A deterministic rule removes one trip per hour from all five stops; an equal-distance tie removes the earlier trip.	Results can depend on which trip is removed.
Bus events precede passenger events at exact ties	Priority 0 is a bus event and priority 1 is a passenger event. The tie convention is explicit and shared by all methods.	A passenger at exactly the bus time misses that bus.
No extra buses are added to clear queues	Demand ends at the period boundary; eligible downstream bus events may finish before backlog is measured.	Final backlog represents the finite study service rather than unlimited later service.
Absent source combinations are not filled with zero	A missing combination is different from an observed zero. The current means use available valid records.	Uneven missing coverage can bias the demand estimates.
Only arrival timestamps change across independent runs	Matched alternatives reuse passenger streams. The independent test uses unused seeds after plans are frozen.	Intervals describe timestamp sampling uncertainty, not input uncertainty or actual operating-day variation.

Note. Sources: current notebooks, m66_core.py, m66_support.py, m66_evaluation.py, and the cited input and arrival-model sources. Each assumption has a corresponding rule or scope boundary in the runnable model.

4.5 Passenger Generation

For each stop-hour cell, the model rounds the nonnegative mean boarding count to the nearest whole number using $\text{floor}(\text{mean} + 0.5)$. The model then places that fixed number of passengers at independent uniform times within the hour. For example, a rounded count of 20 creates exactly 20 arrivals during that hour in every replication, but their times change with the seed. This is a fixed-count uniform model, not a Poisson count model. Replications represent arrival-time uncertainty rather than variation in total daily demand.

The baseline generates 500 selected-stop passengers in the morning and 276 in the evening. The high-demand condition multiplies the unrounded means by 1.40 before rounding, producing 698 and 385 passengers. Rounding means these totals are not exactly 40 percent above the baseline totals. The arrival generator preserves each baseline passenger's ID and time, then generates only the extra passengers from a separate random stream based on the pair (seed, 1). This keeps comparisons matched while avoiding reuse of the baseline arrival draws for the extra passengers.

4.6 Bus Events and Loads

Reference bus events use eligible GTFS trip times. Eligibility and the study hour are determined when the trip reaches Stop 6. A trip entering within the period may reach a downstream stop after 11:00 or 20:00. The

model allows those downstream events to finish but adds no extra buses to clear the queues. This rule preserves whole trips and makes the final backlog meaningful.

The model estimates an upstream net flow from average boardings minus average alightings at Stops 1 through 5. Negative net flow is clipped to zero. The model rounds the hourly target once and divides it among the eligible buses using quotient and remainder. Trip ID order breaks remainder ties. If an hourly target is too large for the buses' combined capacity, the run stops. Reallocating trips changes per-bus starting loads but preserves the total hourly upstream target.

The model distributes each selected stop's hourly alighting target among buses by the same quotient-and-remainder method. The target hour is the trip's Stop 6 entry hour at every modeled stop, even if a downstream event falls into a later clock hour. Assigned alightings are capped by the onboard count at that event. This is an aggregate flow approximation, not a passenger destination model. The high-demand scenario increases selected-stop boardings while keeping upstream flow and alighting targets unchanged.

4.7 Simulation Workflow

Each event contains its time, priority, type, stop sequence, and passenger or trip ID. The engine processes the earliest event first. At an exact-time tie, the bus event has priority 0 and the passenger event priority 1, so the new passenger misses that bus. A passenger event appends the person to the stop queue. A bus event removes actual alightings, boards from the queue, records each boarded passenger's wait, and updates occupancy and queue histories. All scenarios and candidate plans use `m66_core.py` as the single event engine.

The period-end queue is a snapshot at the official boundary after bus events exactly at 11:00 or 20:00. It is recorded before later eligible downstream events. Final backlog is the sum of passengers still waiting after the last eligible event. These outputs may differ. Completed waiting time is boarding time minus queue-entry time. Passengers who never board have no completed wait, so the analysis always reports waiting metrics alongside backlog.

5. AI Design

With the shared simulation defined, the next step is to choose which service decisions AI can change. This simulation project keeps passenger behavior and event rules fixed, screens three search methods under common controls, and uses GA to propose feasible hourly allocations.

5.1 AI Role and Rationale

The AI component is an integer genetic algorithm for service allocation. It changes the hourly number of trips rather than predicting demand or generating passenger behavior. Its input includes stop-hour demand, upstream and alighting targets, travel offsets, hourly bounds, the fixed trip budget, and matched passenger streams. Its output is a five-integer service plan and a candidate schedule. The event engine evaluates that schedule and returns fitness and service-quality checks.

The project team initially considered an AI role that predicted hourly boardings. Following Ms. Roselle's suggestion, the project uses service optimization because the project question asks how changing service affects waiting under specified demand. A predictive model would first need a separate forecasting target and time-based validation design. In the current project, demand remains an explicit scenario input and AI

changes a decision that can be tested directly. Section 11 records this choice among the key project decisions.

Oliveira et al. (2024) use a biased random-key genetic algorithm in public transport optimization. That study supports GA as a relevant family of methods, but this project's implementation is a simpler integer-vector GA with different variables and constraints. This simulation project does not claim to reproduce that study or to match its results.

5.2 Technique Comparison

The technique screen compares GA, simulated annealing (SA), and particle swarm optimization (PSO) on the baseline. Each method receives the same 10 unique-plan budget per run, optimizer seeds 101, 202, and 303, and five matched passenger replications for search fitness. Selected screen plans are evaluated on the larger existing selection set. This is a small technique screen, not the independent test and not a general benchmark of the methods.

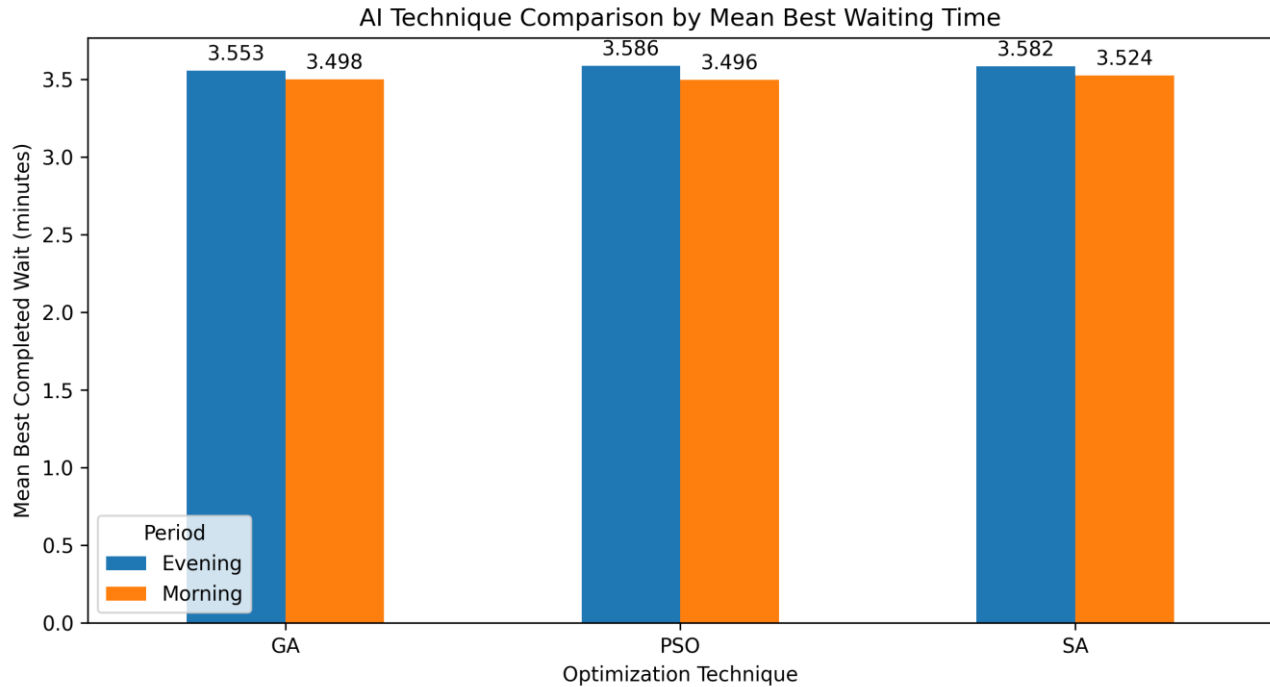
Table 7. Matched baseline technique screen

Method	Morning wait	Evening wait	Equal period average
GA	3.4977	3.5529	3.5253
PSO	3.4956	3.5861	3.5408
SA	3.5245	3.5820	3.5533

Note. Source: ai_technique_summary.csv. Wait is in minutes and is averaged across three optimizer runs after larger-set selection evaluation. The equal period average gives morning and evening the same weight.

GA has the lowest equal-period screen average, 3.5253 minutes, and directly represents integer trip counts. PSO is slightly lower in the morning, 3.4956 versus GA's 3.4977 minutes. These descriptive differences have no paired intervals in Table 7 and support a project-specific method choice, not a general ranking. Section 7 evaluates the selected GA plans against service controls.

Figure 5. Baseline technique screen by period



Note. Source: repository output/figures/03_ai_technique_comparison.png. Bars use the matched small-budget screen and larger-set evaluation. They do not show the final 100-plan GA experiment.

5.3 Decision Variables and Fitness

A candidate is five whole hourly trip counts. Its sum must equal the scenario's period budget. Baseline and high demand allow 5 to 12 trips per hour; reduced service and combined allow 4 to 12. These are experimental bounds, not validated operational limits. Candidate buses use equal midpoint spacing within an hour. For a count of six, Stop 6 times are five, fifteen, twenty-five, thirty-five, forty-five, and fifty-five minutes into that hour. Downstream times add the hour-specific median GTFS offset from Stop 6.

Fitness is the mean completed wait over the five search replications. An invalid plan, nonfinite objective, or mean final backlog above the matched reference receives infinite fitness. The safeguard checks a mean across matched replications; it does not guarantee that every replication or individual passenger improves. The optimizer rejects nonfinite and fractional counts, repairs proposed plans to the budget and bounds, and rebuilds load assignments for the candidate schedule.

5.4 GA Search and Plan Selection

The final GA uses a population of 12 and three optimizer seeds. It begins with the reference allocation and the demand-rule allocation, then adds feasible plans. It evaluates unique plans, keeps the best two scored plans, selects parents through tournaments of up to three, and creates children by choosing each hourly count from either parent with equal probability. It repairs the child and applies a neighboring-plan mutation with probability 0.5. Repeated plans use cached fitness. If the search stalls for five generations, an unseen feasible plan is injected. Each run stops after 100 unique plans.

Search results are rechecked in search-fitness order on the larger selection set, starting at 30 replications under the same precision rule used for the scenario analysis. For each optimizer seed, the selection process

takes the first candidate that passes the unchanged mean backlog safeguard. The selection process then selects the lowest larger-set mean wait among the passing seed candidates. Six reduced-service or combined optimizer runs use candidate ranks 2 to 4 because their search minimum does not pass the larger-set safeguard. The output saves both the search minimum and the passing candidate rank.

The larger-set tables include validation in their filenames, but these results participate in plan selection. Plans are frozen before the independent test. Each optimizer run evaluates 100 unique plans, about 2.6 to 4.2 percent of the 2,380 to 3,900 feasible allocations; evaluated plans can overlap across the three optimizer seeds. Including the demand rule initially does not guarantee better independent-test performance. Search uses five replications, and selection takes the first passing candidate per seed in search-fitness order rather than ranking every plan on the larger set. Both search noise and the backlog constraint can affect selection.

6. Experimental Design

The GA can propose an allocation, but its usefulness depends on a fair comparison. The experiment defines four operating conditions, keeps each service budget explicit, and evaluates published, regularized, demand-rule, and GA plans under matched passenger arrivals. Plan selection and the final test use separate seed sets.

6.1 Operating Scenarios

Table 8. Demand and service conditions

Condition	Demand factor	Morning passengers and trips	Evening passengers and trips
Baseline	1.00	500 passengers / 43 trips	276 passengers / 44 trips
High demand	1.40 before rounding	698 passengers / 43 trips	385 passengers / 44 trips
Reduced service	1.00	500 passengers / 38 trips	276 passengers / 39 trips
Combined	1.40 before rounding	698 passengers / 38 trips	385 passengers / 39 trips

Note. Sources: scenario_passenger_demand_validation.csv and scenario_hourly_trip_counts.csv. Trips are counted at Stop 6. The factor changes selected-stop boardings only.

Reduced service removes one whole trip per study hour. The reduced-service rule removes the eligible trip with a Stop 6 time closest to the half-hour, choosing the earlier trip for a tie. The model removes that trip at every selected stop and rebuilds upstream and alighting allocations. This gives 38 morning and 39 evening trips. It is a controlled service-loss experiment, not a modeled incident or a forecast. The 40 percent demand increase is also a chosen stress level.

6.2 Service Plans and Controls

For each condition, the evaluation compares four service methods. The published reference uses eligible GTFS times or the defined reduced-service subset. The regularized reference preserves its hourly trip counts but uses the same midpoint-spacing and median-offset rules as GA. The demand rule distributes trips in proportion to the hourly sum of selected-stop boarding demand, repaired to the same budget and bounds. GA uses the selected hourly allocation with those same candidate timing rules. The standalone demand-rule control is evaluated without the GA search or larger-set backlog rejection rule. It remains a transparent allocation comparator, rather than a plan certified by the selection safeguard.

GA versus published reference includes both timing regularization and hourly reallocation. GA versus regularized reference isolates the allocation contrast under common timing rules. GA versus demand rule tests whether searching improves on a simpler allocation method.

6.3 Evaluation Metrics

Table 9. Reported simulation and optimization measures

Measure	Definition and interpretation
Mean completed wait	Mean minutes from arrival to boarding among passengers who board. Primary GA objective.
P95 completed wait	95th percentile of completed waits within a replication. Summary tables average these replication percentiles.
Peak queue	Largest queue at a single stop during a replication. Summary tables average replication peaks, not simultaneous network totals.
Period-end queue	Sum of the five queues at the official period boundary.
Final backlog	Sum still waiting after all eligible events. Used with completed wait and as a search and selection safeguard.
Mean bus occupancy	Average post-event onboard count across recorded bus-stop events. Not a route-distance or time-weighted load.
Maximum bus occupancy	Largest recorded post-event load; checked against the assumed capacity.
Search progress and seed variation	Unique plans evaluated, best fitness history, and the selected plan and performance from each optimizer seed.

When no passenger boards, completed-wait metrics are missing rather than zero. When no bus event exists, occupancy summaries are missing. All reported current comparisons require finite metrics and complete matched keys.

6.4 Replications and Statistical Analysis

The evaluation uses matched passenger streams for all methods within a replication. Baseline and high demand also share the baseline passenger arrivals, while high demand adds extra passengers. This matching reduces noise in differences caused by service changes. The scenario and selection runs start at 30 replications. For mean completed wait and peak queue, a 95 percent Student t interval must have half-width at most 1 unit when the mean is below 10, or at most 10 percent of the mean otherwise. If needed, the code adds 10 replications at a time up to 50. All current scenario and selected-plan checks pass at 30.

Law (2020) explains independent replications, Student t intervals, and how more replications can improve interval precision. The project's 30-to-50 replication limit and precision thresholds are project-specific choices. The uncertainty covered by these intervals is defined in Section 2.2.

For the independent test, the evaluation generates 30 unused passenger seeds from `SeedSequence([2026, 90902])` and assigns replication IDs 1001 to 1030. The seed check confirms that the seeds did not occur in selection inputs. The plans remain fixed throughout this test. These are unused simulated arrival realizations,

not new observed operating days. There are 240 reference replication rows and 720 candidate rows across four conditions, two periods, and three candidate methods.

For a comparison, the analysis subtracts the reference result from the candidate result for each matched replication and calculates the mean difference and a two-sided 95 percent Student t interval with 29 degrees of freedom. A negative difference means the candidate has a lower metric. An interval entirely below zero supports a reduction for that comparison; one entirely above zero supports an increase; one including zero is inconclusive. Intervals are pointwise and are not adjusted for the many metrics and stops tested. The analysis reports the actual intervals and effect sizes instead of claiming simultaneous certainty. Stop comparisons remain exploratory. No non-inferiority margin was specified, and none is introduced after inspecting the results. An interval containing zero is not evidence that deterioration has been ruled out.

Model checks confirmed passenger conservation, nonnegative waiting times, occupancy within the assumed capacity, fixed trip budgets, and matched replication seeds. These checks verify implementation consistency but do not establish accuracy against observed passenger queues.

7. Results and Discussion

The results begin with reference stress effects and the frozen plans, then distinguish published-schedule changes from allocation performance. Comparisons with the regularized reference and demand rule provide the main evidence about hourly allocation. Queue, occupancy, and stop results show the accompanying tradeoffs.

7.1 Reference Scenario Results

Table 10. Independent test performance of the reference service

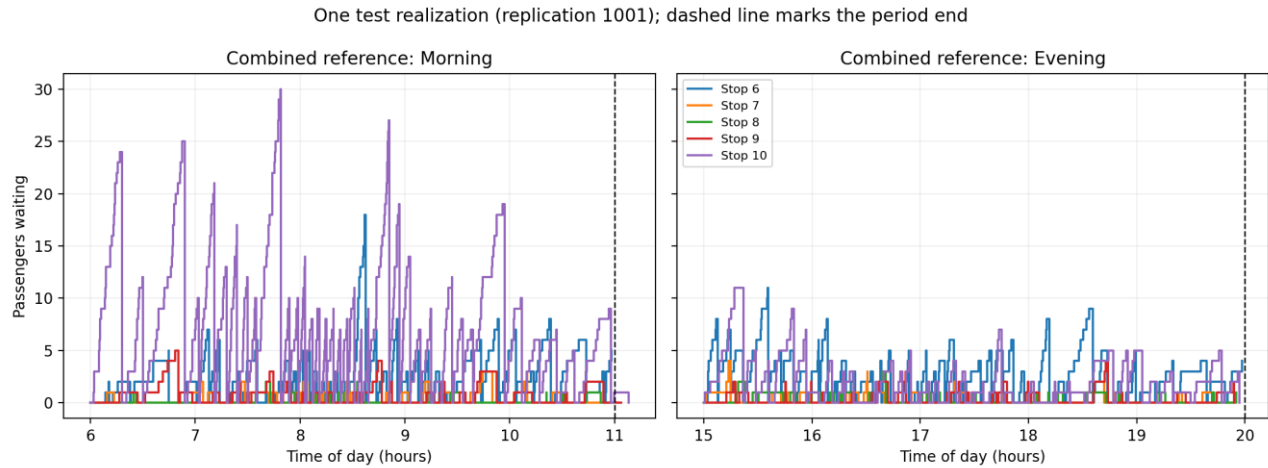
Condition	Period	Mean wait	Mean P95	Mean peak	Mean backlog
Baseline	Morning	3.646	9.063	17.80	1.63
Baseline	Evening	3.593	7.802	9.83	3.57
High Demand	Morning	3.634	9.035	23.07	2.60
High Demand	Evening	3.618	7.827	14.00	5.23
Reduced Service	Morning	4.763	13.920	23.73	1.63
Reduced Service	Evening	4.324	11.657	10.50	3.57
Combined	Morning	4.733	13.903	32.00	2.60
Combined	Evening	4.339	11.838	14.43	5.23

Note. Source: review_holdout_reference_replications.csv, 30 replications per row. Wait values are minutes; queue and backlog values are passengers. Mean P95 and mean peak are averages of replication statistics.

Removing one trip per hour increases reference mean wait from 3.646 to 4.763 minutes in the morning and from 3.593 to 4.324 in the evening. The independent paired increases are 1.118 minutes [1.069, 1.166] and 0.731 minutes [0.693, 0.769], respectively. The combined condition has similar completed mean waits but larger queues than reduced service alone. The largest mean reference peak is 32.00 passengers under combined morning conditions.

High demand increases reference peak queues and backlog, but it does not clearly increase mean completed wait. Its paired mean-wait differences from baseline are -0.012 minutes [-0.033, 0.010] in the morning and 0.025 minutes [-0.007, 0.057] in the evening. Both intervals contain zero. With deterministic service and mostly nonbinding capacity, extra passengers can enlarge queues without strongly changing the wait distribution among people who board.

Figure 6. Combined reference queue behavior at the five stops



Note. Source: repository output/figures/review_reference_queue_combined.png, produced by plot_results.py. One independent realization, replication 1001, is shown for each period. Dashed lines mark period boundaries. These traces illustrate events and are not replication averages.

7.2 Frozen GA Plans

Table 11. GA allocations frozen before the independent test

Condition	Period	Hourly trip counts	Total
Baseline	Morning	(8, 9, 12, 8, 6)	43
Baseline	Evening	(10, 10, 10, 8, 6)	44
High Demand	Morning	(7, 12, 11, 7, 6)	43
High Demand	Evening	(11, 8, 10, 8, 7)	44
Reduced Service	Morning	(7, 9, 10, 7, 5)	38
Reduced Service	Evening	(10, 7, 10, 6, 6)	39
Combined	Morning	(7, 9, 10, 7, 5)	38
Combined	Evening	(10, 7, 10, 8, 4)	39

Note. Source: review_frozen_plans.csv. Morning vector positions are hours 06, 07, 08, 09, and 10. Evening positions are 15, 16, 17, 18, and 19. Reference baseline counts are (5, 12, 12, 8, 6) and (8, 12, 11, 8, 5); reduced counts subtract one from each position.

The selected plans preserve their budgets. The same morning allocation is selected for reduced service and combined conditions. This is a result of the limited search and selection procedure, not a rule forcing the plans to be equal. Candidate schedules rebuild per-bus starting loads and alighting assignments when hourly

counts change. For example, baseline morning shifts three trips from the second study hour into the first, while keeping the period budget fixed.

7.3 Independent Waiting Time Results

Table 12. GA mean completed wait compared with the published reference

Condition	Period	Reference	GA	Reduction percent	95 percent CI for difference
Baseline	Morning	3.646	3.499	4.03	[-0.200, -0.094]
Baseline	Evening	3.593	3.483	3.07	[-0.174, -0.046]
High Demand	Morning	3.634	3.489	4.00	[-0.184, -0.107]
High Demand	Evening	3.618	3.528	2.50	[-0.134, -0.046]
Reduced Service	Morning	4.763	3.903	18.06	[-0.938, -0.783]
Reduced Service	Evening	4.324	4.012	7.21	[-0.382, -0.241]
Combined	Morning	4.733	3.904	17.52	[-0.885, -0.774]
Combined	Evening	4.339	3.994	7.93	[-0.401, -0.288]

Note. Source: review_holdout_paired_comparisons.csv, Mean Completed Wait and GA minus Published Reference rows. Wait and intervals are in minutes. Differences use GA minus reference; all eight intervals are below zero. Display values are rounded after computation.

GA lowers mean completed wait against the published reference in all eight condition-period comparisons. The largest relative reduction is reduced-service morning, from 4.763 to 3.903 minutes, a reduction of 18.06 percent or about 51.6 seconds per completed wait. The combined morning reduction is 17.52 percent. Baseline reductions are smaller, about 8.8 seconds in the morning and 6.6 seconds in the evening. These are average effects for passengers who board, not a guaranteed saving for every passenger.

This published-reference gain combines timing regularization and hourly reallocation. Section 7.4 separates those contrasts and compares GA with the demand rule; these are the main tests of allocation performance.

7.4 Timing and Allocation Effects

Table 13 separates timing from allocation. Five timing intervals support reductions; baseline morning, baseline evening, and high-demand evening are inconclusive. In baseline morning, the allocation contrast is 0.113 minutes and the timing contrast is 0.034, compared with a total reduction of 0.147. Most of this descriptive mean reduction comes from the allocation contrast, while the timing interval includes zero. Reduced-service morning differs: timing accounts for 0.658 minutes and allocation for 0.203 before rounding. The joint outcomes in Table 14 and the paired allocation tests that follow show why the control matters.

Table 13. Simulated timing and allocation contrasts

Condition	Period	Timing reduction	95 percent CI for timing	Allocation reduction
Baseline	Morning	0.034	[-0.015, 0.083]	0.113
Baseline	Evening	0.001	[-0.061, 0.063]	0.109
High Demand	Morning	0.044	[0.002, 0.085]	0.101

Condition	Period	Timing reduction	95 percent CI for timing	Allocation reduction
High Demand	Evening	0.002	[-0.046, 0.049]	0.089
Reduced Service	Morning	0.658	[0.586, 0.730]	0.203
Reduced Service	Evening	0.211	[0.140, 0.282]	0.100
Combined	Morning	0.627	[0.578, 0.677]	0.202
Combined	Evening	0.232	[0.170, 0.294]	0.112

Note. Source: review_holdout_paired_comparisons.csv, Mean Completed Wait rows, 30 matched replications. Values are minutes. Positive values indicate lower mean completed wait under regularized service for the timing contrast and under GA for the allocation contrast. Timing is published minus regularized; allocation is regularized minus GA. Timing interval endpoints reverse the saved regularized-minus-published contrast. The reductions sum to the published-to-GA gain before rounding. Allocation intervals are reported in Table 15. Differences were calculated before rounding the displayed values.

Table 14. Mean completed wait and final backlog by service method

Condition	Period	Published	Regularized	Demand rule	GA
Baseline	Morning	3.646 / 1.633	3.611 / 1.200	3.536 / 1.567	3.499 / 1.200
Baseline	Evening	3.593 / 3.567	3.592 / 0.933	3.504 / 0.933	3.483 / 0.767
High Demand	Morning	3.634 / 2.600	3.590 / 1.967	3.523 / 2.533	3.489 / 1.967
High Demand	Evening	3.618 / 5.233	3.616 / 1.733	3.520 / 1.733	3.528 / 1.067
Reduced Service	Morning	4.763 / 1.633	4.105 / 1.567	4.049 / 2.100	3.903 / 1.567
Reduced Service	Evening	4.324 / 3.567	4.113 / 1.433	3.926 / 1.433	4.012 / 0.767
Combined	Morning	4.733 / 2.600	4.106 / 2.533	4.027 / 3.233	3.904 / 2.533
Combined	Evening	4.339 / 5.233	4.107 / 2.400	3.942 / 2.400	3.994 / 2.400

Note. Sources: review_holdout_reference_replications.csv and review_holdout_candidate_replications.csv, 30 matched unused-seed replications. Each cell is mean completed wait in minutes / mean final backlog in passengers. Joint display shows the two outcomes; it does not replace the paired intervals in the following tables or imply that the methods satisfy identical selection constraints. Differences were calculated before rounding the displayed values.

Table 15. GA mean completed wait compared with the regularized reference

Condition	Period	Regularized	GA	Difference	95 percent CI
Baseline	Morning	3.611	3.499	-0.113	[-0.155, -0.070]
Baseline	Evening	3.592	3.483	-0.109	[-0.166, -0.052]
High Demand	Morning	3.590	3.489	-0.101	[-0.136, -0.067]
High Demand	Evening	3.616	3.528	-0.089	[-0.128, -0.049]
Reduced Service	Morning	4.105	3.903	-0.203	[-0.251, -0.155]
Reduced Service	Evening	4.113	4.012	-0.100	[-0.170, -0.031]
Combined	Morning	4.106	3.904	-0.202	[-0.238, -0.166]
Combined	Evening	4.107	3.994	-0.112	[-0.156, -0.069]

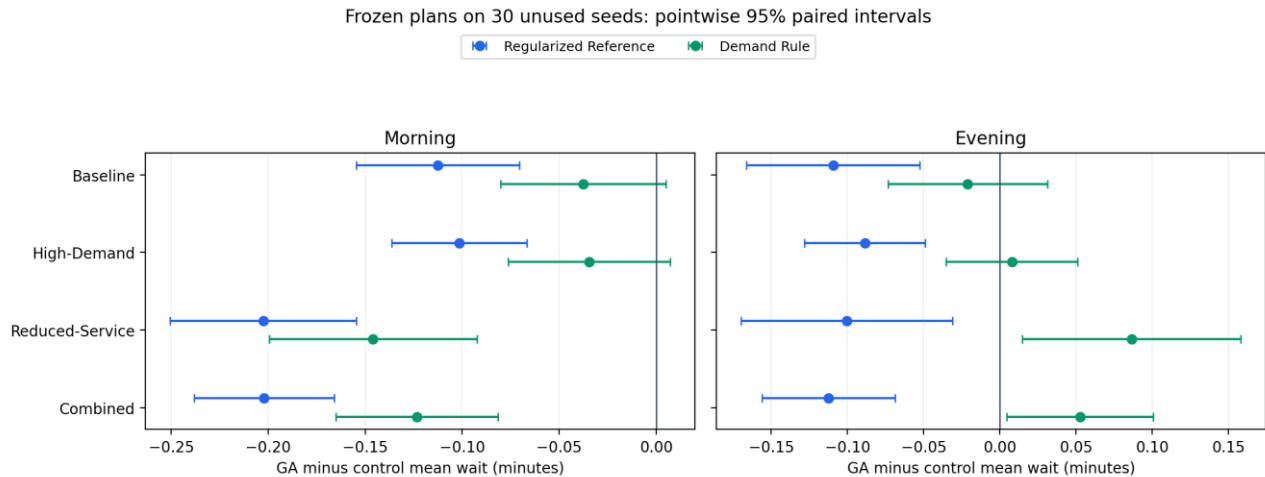
Note. Source: review_holdout_paired_comparisons.csv, Mean Completed Wait and GA minus Regularized Reference rows. Values are minutes from 30 matched unused-seed replications. Negative differences favor GA. Intervals are pointwise. Differences were calculated before rounding the displayed values.

Table 16. GA mean completed wait compared with the demand rule

Condition	Period	Difference	95 percent CI	Interpretation
Baseline	Morning	-0.038	[-0.080, 0.005]	Inconclusive
Baseline	Evening	-0.021	[-0.073, 0.031]	Inconclusive
High Demand	Morning	-0.035	[-0.076, 0.007]	Inconclusive
High Demand	Evening	0.008	[-0.035, 0.051]	Inconclusive
Reduced Service	Morning	-0.146	[-0.199, -0.092]	Lower wait
Reduced Service	Evening	0.086	[0.015, 0.158]	Higher wait
Combined	Morning	-0.123	[-0.165, -0.082]	Lower wait
Combined	Evening	0.053	[0.005, 0.101]	Higher wait

Note. Source: independent paired comparison table, GA minus Demand Rule and Mean Completed Wait rows. Differences and intervals are minutes. Lower and higher refer to GA relative to the demand rule. Differences were calculated before rounding the displayed values.

Figure 7. Independent paired GA mean wait differences against allocation controls



Note. Source: repository output/figures/review_holdout_wait_differences.png, produced by plot_results.py from the independent paired comparison CSV. Points are matched mean differences and lines are pointwise 95 percent intervals from 30 replications. Negative values favor GA. Each period panel has its own scale.

GA does not dominate the demand rule. It has supported improvements in reduced-service morning and combined morning. It has supported increases in reduced-service evening, about 5.2 seconds, and combined evening, about 3.2 seconds. The other four mean-wait comparisons are inconclusive. A practical use should retain the demand rule as a candidate method and compare its backlog and stop results, especially for evening service. The demand rule has higher final backlog than published service in reduced-service morning by 0.467 passengers [0.232, 0.701] and combined morning by 0.633 [0.365, 0.902]. These independent-test increases show why its mean-wait result must be read with backlog. They do not describe whether a candidate passed the separate GA selection sets.

7.5 Tail Waiting Time and Queues

GA lowers the mean of replication P95 completed waits against the published reference in all eight comparisons. The largest decrease is 5.60 minutes in combined morning, where the reference mean P95 is 13.903 and GA is 8.305 minutes. This describes the upper part of completed waits; it does not include people who never board. Appendix A gives all P95, queue, and backlog differences and intervals.

For mean peak queue, six of eight comparisons against the published reference support reductions; baseline and high-demand morning are inconclusive. For final backlog, six support reductions; reduced-service and combined morning are inconclusive. Section 6.4 defines the interval interpretation.

Mean period-end queue is higher under GA in reduced-service morning (3.50 to 4.30 passengers), combined morning (5.13 to 6.20), and combined evening (5.23 to 5.93). These are differences between reported means, not claims of statistically supported increases. Eligible downstream buses continue after the official boundary, so a higher period-end queue can coexist with lower final backlog and shorter completed waits. Improvement in waiting does not imply improvement in every queue measure.

Across all 4,800 stop-replication records, final backlog equals arrivals at or after the last eligible bus. Exact-time ties count because bus events take priority. No earlier-arriving passenger remains in final backlog, although temporary boarding denials are not counted by this diagnostic. The measure describes cutoff service coverage in these tests. Appendix F gives stop-specific times and counts.

The saved paired comparisons also separate backlog effects from published timing. Against the regularized reference, GA has exactly the same final-backlog outcome in every matched morning replication and in combined evening. Mean backlog is lower in baseline evening (0.933 to 0.767), high-demand evening (1.733 to 1.067), and reduced-service evening (1.433 to 0.767); each corresponding paired interval is below zero. Against the demand rule, seven final-backlog intervals are below zero and combined evening is exactly equal. Source: review_holdout_paired_comparisons.csv, Final Backlog rows. These comparisons use the same midpoint-spacing rule, but their last-bus times can still differ when hourly trip counts change.

Table 17. Simulated maximum bus occupancy

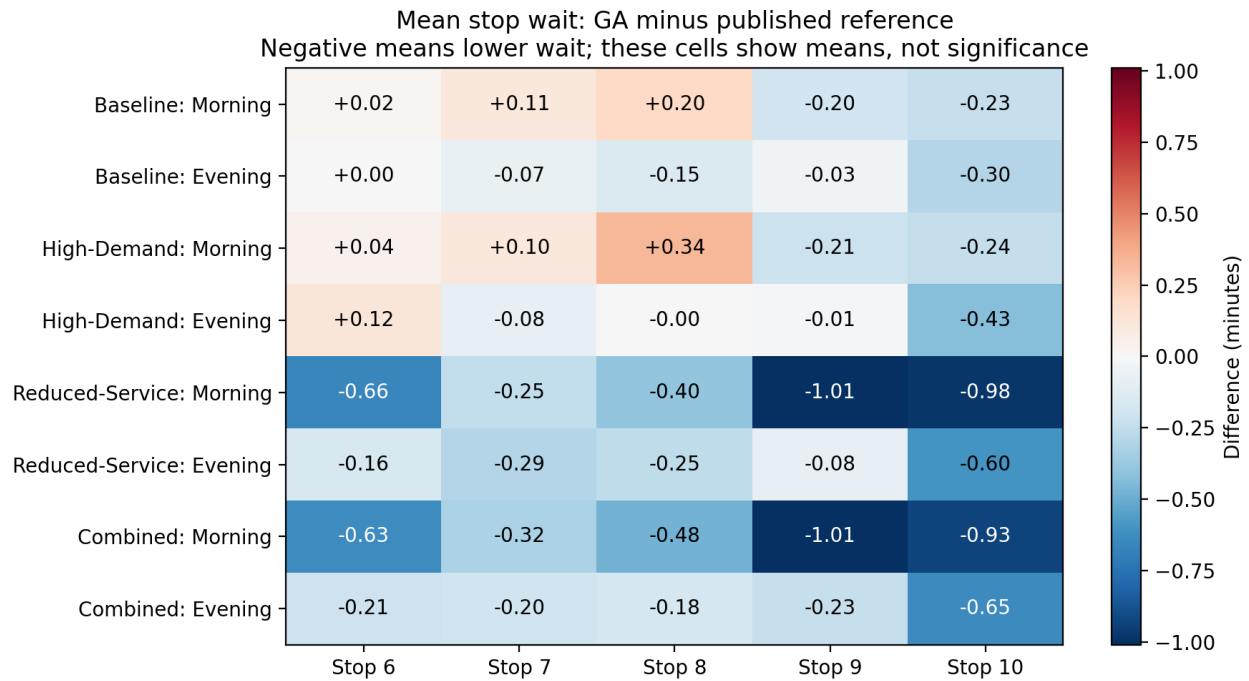
Condition and period	Published	Regularized	GA	Demand rule
Baseline Morning	29.53 (33)	29.60 (37)	30.87 (35)	28.03 (35)
Baseline Evening	24.30 (28)	24.17 (28)	20.07 (26)	19.47 (22)
High Demand Morning	36.27 (42)	37.30 (46)	35.57 (39)	34.80 (41)
High Demand Evening	29.13 (38)	26.70 (30)	23.60 (27)	21.90 (25)
Reduced Service Morning	41.47 (48)	33.60 (39)	32.07 (35)	32.03 (39)
Reduced Service Evening	27.77 (32)	26.60 (29)	24.27 (29)	21.03 (26)
Combined Morning	53.20 (60)	42.53 (52)	39.83 (44)	39.43 (50)
Combined Evening	32.03 (38)	29.40 (33)	26.27 (31)	23.30 (28)

Note. Sources: review_holdout_reference_replications.csv and review_holdout_candidate_replications.csv, Maximum Bus Occupancy. Each cell shows the mean of 30 replication maxima, followed in parentheses by the largest recorded maximum. Values are passengers on board after selected-stop bus events; assumed capacity is 60. These are simulated loads, not measured vehicle occupancy.

Combined morning published service reaches the assumed capacity of 60 in at least one replication. All other condition-period-method groups have recorded maxima below 60. Reaching capacity does not by itself count denied passengers. This supports inspecting the combined morning capacity condition while leaving denial-event reporting as the extension in Section 9.

7.6 Stop Tradeoffs and Search Evidence

Figure 8. Independent stop mean wait differences under GA



Note. Source: repository output/figures/review_holdout_stop_wait_tradeoffs.png. Cells show mean GA-minus-published differences in minutes across all conditions and periods. Cell colors show direction and magnitude, not significance. The paired stop-comparison CSV supplies the intervals discussed in the text.

The analysis examines 40 GA-versus-published-reference stop mean-wait comparisons: four conditions, two periods, and five stops. Seventeen intervals are entirely below zero, 21 include zero, and two high-demand intervals are entirely above zero even though the period mean improves. Morning Stop 8 increases by 0.338 minutes [0.058, 0.618], about 20.3 seconds. Evening Stop 6 increases by 0.118 minutes [0.065, 0.171], about 7.1 seconds. These exploratory pointwise intervals are not adjusted for multiple testing. Stop averages give equal weight to their replications, while the period metric summarizes all completed passengers within each replication. A period improvement can coexist with a local worsening.

Table 18. Stop mean-wait comparisons against regularized service

Condition	Period	Lower	Inconclusive	Higher	Supported increases
Baseline	Morning	2	3	0	None
Baseline	Evening	2	3	0	None
High Demand	Morning	2	3	0	None
High Demand	Evening	1	4	0	None
Reduced Service	Morning	2	2	1	Stop 7: +0.291 [0.089, 0.493]
Reduced Service	Evening	1	4	0	None
Combined	Morning	2	2	1	Stop 7: +0.304 [0.145, 0.463]
Combined	Evening	2	3	0	None

Note. Source: paired calculations from `review_holdout_candidate_stops.csv`, using 30 seed-matched pairs per stop and the Student t procedure in Section 6.4. Difference is GA minus regularized reference. Counts cover five stops per row; intervals are exploratory and pointwise. Differences were calculated before rounding the displayed values. Differences and 95 percent confidence intervals are in minutes.

Using the common midpoint-spacing control changes the local interpretation. Fourteen stop intervals support reductions, 24 are inconclusive, and two support increases. Both increases occur at Stop 7: reduced-service morning rises by 0.291 minutes [0.089, 0.493], about 17.5 seconds, and combined morning rises by 0.304 [0.145, 0.463], about 18.3 seconds. The two increases against published service do not remain supported against regularized service. The allocation comparison identifies a different local tradeoff, even though every period-level GA mean is lower than the regularized reference.

All 24 scenario-period-seed searches reached the 100 unique-plan budget, and their saved best fitness was nonincreasing in `ga_search_history.csv`. These scores use five search replications. Selection can replace a search minimum when its mean backlog fails on the larger set; final service comparisons use frozen plans and the independent test.

8. Ethical Considerations

The public records are aggregate stop-hour counts, and generated passenger IDs are synthetic labels. The project does not identify riders or infer demographic characteristics. Stop-level results in Section 7.6 guide checks for uneven service effects; benefits for specific passenger groups require further evidence.

NIST's AI Risk Management Framework describes managing risk in the intended context of use (Tabassi, 2023). This project traces results to inputs, reports uncertainty, and retains unfavorable comparisons. The recommended operational checks are listed in Section 9.

9. Recommendations

The following future work would extend the completed service comparison.

1. **Operational Validation.** Collect passenger queue-entry and boarding times, bus arrivals, queues, and onboard loads for matching service dates. Compare observed and simulated outcomes to test the arrival and load rules. Matching demand and schedule dates would also address the current cross-year mismatch.
2. **Sensitivity Analysis.** Vary capacity, initial queues, arrival patterns, upstream loads, and alighting assignments in matched experiments to test whether plan advantages persist. Identify hypothetical changes separately from observed operating conditions.
3. **Route Operational Feasibility.** Extend the model to vehicle circulation, recovery time, fleet availability, and driver duties. Service planners should check whether candidate timetables can be delivered with available resources before a field trial (Gkiotsalitis, 2020).
4. **Search Benchmarking.** Compare GA and random search with the same feasible space, unique-plan budget, passenger replications, safeguards, and selection procedure. Exhaustive enumeration could establish the best plan for a declared finite simulation objective. Freeze plans before a fresh independent test. These experiments would evaluate search efficiency and optimality. Report revised search or selection procedures as separate experiments and retain unfavorable results.

5. **Capacity and Boundary Diagnostics.** Count unique passengers denied boarding and total denial events. Compare schedules with matching final bus times to separate cutoff coverage from other effects, while documenting the changed timing rule and preserving trip budgets. These extensions would clarify capacity and boundary mechanisms beyond the saved last-bus decomposition.
6. **Stop Fairness and Statistical Inference.** Evaluate worst-stop outcomes and explicit minimum-service standards before operational use. A new confirmatory study should define primary comparisons, practical non-inferiority margins, and any multiple-testing adjustment before examining new results. Current stop comparisons remain exploratory.

10. Member Contributions

The project team records the responsibilities and completed work below. Notebook and module paths identify the associated artifacts, while Appendix E links to relevant commit diffs. Current files also contain later integration changes.

Table 19. Member contributions supported by GitHub

Member	Responsibilities	Work completed	GitHub evidence
Elijah Crisehea M. Nollen	Dataset analysis, baseline simulation, integration, QA, and documentation	Completed dataset cleaning, quality checks, and descriptive analysis; developed the baseline simulation and technique screen; integrated shared code, scenario and GA work, and current documentation.	code/m66_dataset_analysis.ipynb; code/m66_baseline_simulation.ipynb. Integration diff 7fb5985b; shared modules, run_all.py, requirements, README, and technical documentation.
Ma. Sofia Anne C. Revilla	GA development and scenario integration	Integrated high-demand and reduced-service contexts into the GA; revised the notebook with paired intervals, precision checks, and the larger-set backlog safeguard.	Commits a8f7de0c and 89ade5ad. Current m66_ga_optimization.ipynb and GA output tables.
Karol Joy M. Ronquillo	Scenario simulation and scenario evidence	Rewrote the scenario notebook; maintained demand and service scenario inputs; produced and revised scenario results and comparison figures.	Commits 5625be8b and 415b2b5f. Current m66_scenario_simulation.ipynb and scenario tables and figures.

Note. Repository: <https://github.com/elijahnollen/m66-passenger-queue-simulation>. Full commit links are provided in Appendix E.

11. Project Decisions and AI Disclosure

This section records the major choices in the M66 simulation and how AI tools supported its development. AI assistance included reviewing assumptions, suggesting Python code, explaining scenario checks, comparing optimization methods, and improving the report. The final choices are based on the implemented model and its saved outputs.

11.1 Key Project Decisions

Table 20. Major choices in the current project

Decision	Rationale	AI influence	Final justification
Use AI for service allocation rather than demand prediction	The main question is whether changing service improves waiting under stated demand.	Partial. The Gemini writing chat includes an earlier prediction role; the Claude GA review concerns optimization.	Following Ms. Roselle's suggestion, the project changed from demand prediction to service allocation. Recorded boardings supply demand; GA proposes hourly counts within a fixed budget. Section 5.1 explains the role.
Use frozen ridership and static GTFS inputs	The model needs compatible stop-hour activity and a reproducible timetable.	AI explained the dataset roles and suggested real-time data and other approaches. It also assisted data inspection.	Use the frozen 2025 ridership and 2026 GTFS pattern, with source-checked physical stop mappings. Bus Time is not used. Section 3 documents the inputs.
Keep one shared event engine	The service methods need identical queue, boarding, capacity, and boundary rules.	AI helped separate data preparation from event processing. It supplied event-engine coding examples. It identified unclear state and timing rules.	All scenarios and service plans use <code>m66_core.py</code> with the rules in Section 4.
Keep fixed service budgets and explicit hourly bounds	Candidate plans must redistribute trips within the stated experiment.	AI compared 5 to 12, 4 to 11, and 4 to 12. Scenario guidance also explained integer and budget checks.	Keep each reference count vector inside the feasible space. The budgets and bounds are specified in Section 5.3; they are experimental constraints.
Define reduced service as one whole trip removed per hour	A repeatable removal rule isolates service loss.	AI suggested hourly, midpoint, tie-break, and cross-stop checks for trip removal.	Apply the same deterministic removal rule across all five stops, as specified in Section 6.1.
Preserve baseline passengers when demand increases	Matched arrivals help compare service changes with less unrelated noise.	AI explained multiplying before rounding, retaining baseline arrivals, and generating only additional passengers.	Retain baseline passenger IDs and arrival times and generate only the additional arrivals. Section 4.5 states the rounding and stream rules.
Keep upstream and alighting targets unchanged	The demand increase is limited to boardings at the selected stops.	AI supported the combined-scenario load treatment. Claude clarified that hourly upstream flow stays fixed while per-bus loads change.	Rebuild per-bus assignments while preserving hourly targets. Section 4.6 defines assignment and occupancy caps.

Decision	Rationale	AI influence	Final justification
Separate search, selection, and independent evaluation	A plan can fit the passenger streams used to choose it.	Partial. Scenario guidance discussed small search sets and larger-set checks; AI also raised search noise.	Use larger-set checks for selection, then freeze plans before the unused-seed test. Sections 5.4 and 6.4 distinguish these stages.
Compare GA with timing and demand controls	A published-schedule comparison combines spacing and allocation changes.	Partial. AI proposed the evenly spaced reference control and paired intervals.	Use the four matched methods defined in Section 6.2 and retain unfavorable results.
Report backlog and stop-level effects	Completed waiting time excludes people who never board, and an average can hide local losses.	AI questioned small backlog margins, hidden demand, and whether left-behind tracking solves demand undercount.	Use the selection safeguard in Section 5.3 and report independent backlog and local tradeoffs in Sections 7.5 and 7.6.

11.2 AI Usage Summary

Gemini supported data preparation, Python implementation, passenger-arrival generation, scenario checks, and discussion of hourly trip bounds. Claude explained dataset choices and alternatives, reviewed modeling assumptions and GA consistency, and suggested paired comparisons and a regularized reference control. ChatGPT helped reorganize the simulation explanation into preparation and event processing.

Two examples show how Modified advice was adapted. Gemini’s time-conversion guidance was applied to the actual Date and Hour columns using `pd.to_datetime`, then `.dt.hour.isin(study_hours)` selects hours 6 to 10 and 15 to 19. Generic stop mappings were replaced by checks using route, direction, physical stop names, and sequence. For example, Table 1 connects historical Stop 8 ID 404994 with frozen GTFS ID 400027 rather than assuming the IDs match. These changes align the examples with the project inputs. Poisson-count examples and AI-generated numerical results were not adopted.

11.3 AI Prompt Log

The log separates purposes within shared conversations. Repeated links identify the same conversation used for another purpose. Prompt summaries are not verbatim quotations. For the Claude draft review whose original prompt and attachments are hidden, entries describe only the visible response. Each decision explains how the advice was checked, adapted, or excluded from the final model.

Table 21. Significant AI interactions and group evaluation

AI tool	Purpose, prompt summary, and link	Evaluation	Justification
Claude	Dataset roles. Prompt summary: Explain the public MTA datasets used for the five-stop queue simulation and suggest alternative approaches. Open Claude conversation	Modified	Used the distinction between stop-hour passenger activity and static GTFS service. Recorded boardings are a demand proxy; they do not reveal every person who wanted to travel.
Claude	Alternative data and methods. Prompt summary: Explain the public MTA datasets used for the five-stop queue simulation and suggest alternative approaches. Open Claude conversation	Rejected	Real-time feed collection, pre-built tools, and theoretical queueing examples were alternatives. Bus Time is not an input to the finished simulation. Feed claims and textbook queue labels were not treated as verified project facts.
Gemini	Problem statement and objectives. Prompt summary: Check the draft for clarity and grammar without changing its meaning. Open Gemini conversation	Modified	Used writing advice while updating the final objectives to service allocation, completed waiting, queues, backlog, and occupancy. The earlier boarding-prediction role does not describe the finished project.
Gemini	Date, hour, and period preparation. Prompt summary: Convert date and hour variables, prepare time-based filtering, and create morning and evening categories. Open Gemini conversation	Modified	Adapted pandas examples to the actual columns. Weekdays and hour starts 6 to 10 and 15 to 19 represent the two five-hour windows.

AI tool	Purpose, prompt summary, and link	Evaluation	Justification
Gemini	Zero-trip data issues. Prompt summary: Check records with zero trips and positive boardings or alightings. Open Gemini conversation	Modified	Used the suggested checks to identify conflicting activity records. Cleaning rules and excluded observations must follow the source data rather than a generic example.
Gemini	Missing combinations. Prompt summary: Check expected weekday, study-hour, and selected-stop combinations and inspect dates with missing records. Open Gemini conversation	Modified	Used grid and date checks to distinguish incomplete coverage from observed zero activity. Missing combinations are not automatically evidence of zero demand.
Gemini	Saved inputs and summaries. Prompt summary: Save the cleaned dataset and calculate stop-period and period-level boarding, alighting, and trip summaries. Open Gemini conversation	Modified	Adapted grouping and export code to repository filenames and definitions. Stop-level means and period totals use their stated aggregation rules.
Gemini	Data figures. Prompt summary: Create a grouped boarding chart and a stop-hour heatmap. Open Gemini conversation	Modified	Used plotting guidance with the actual data. Report figures come from repository analysis and simulation outputs; illustrative AI examples are not reported as measured values.
Gemini	GTFS pattern selection. Prompt summary: Inspect eastbound route, service, headsign, and shape information and map stop sequences 6 to 10. Open Gemini conversation	Modified	Verified the retained full-route pattern and physical stops against the source files. The example stop-name mapping and a broad direction or headsign filter were not accepted as proof of the correct corridor.
Gemini	Time conversion and lookups. Prompt summary: Convert HH:MM:SS to minutes and build multi-column dictionaries and grouped time lists. Open Gemini conversation	Modified	Adapted pandas lookups and time conversions to the simulation keys and units. These were coding examples, not a complete verified event engine.
Gemini	Reproducible passenger generation. Prompt summary: Use fixed random seeds, uniform hourly arrival times, and SeedSequence streams. Open Gemini conversation	Modified	Used reproducible stream concepts and uniform arrival times for fixed rounded stop-hour counts. Counts and timestamps are checked separately.
Gemini	Poisson passenger counts. Prompt summary: Sample a Poisson distribution using an expected passenger count. Open Gemini conversation	Rejected	The finished model fixes the rounded number of passengers per stop-hour and randomizes their times. It does not sample hourly passenger counts from a Poisson distribution.

AI tool	Purpose, prompt summary, and link	Evaluation	Justification
Gemini	Event queue and state. Prompt summary: Use heapq for chronological events, same-time priorities, copied queues, and separate period state. Open Gemini conversation	Modified	Applied explicit tuple priorities and fresh replication state. The final engine processes buses before passenger arrivals at equal times; a generic insertion-order example does not define the final tie rule.
Gemini	FIFO boarding and capacity. Prompt summary: Implement FIFO queues, passenger alighting, available capacity, and a chronological event loop. Open Gemini conversation	Modified	Adapted the examples to alighting before boarding, a 60-passenger experimental limit, and carried bus occupancy. Passengers who cannot board remain queued.
Gemini	Integer flow assignments. Prompt summary: Round hourly flows and allocate integer occupancy and alighting counts by quotient and remainder. Open Gemini conversation	Modified	Used the project rounding rule and deterministic assignments that preserve hourly targets. Generic clipping and rounding examples do not justify discarding upstream flow. Realized alighting is limited by onboard passengers.
Gemini	Validation and replication checks. Prompt summary: Check passenger conservation, nonnegative waiting, occupancy limits, generated counts, and reusable-function reproduction. Open Gemini conversation	Modified	Used these checks to compare inputs and outputs and keep the shared engine consistent. Placeholder example results are not evidence that the real simulation passed.
Gemini	Metrics and confidence intervals. Prompt summary: Calculate simulation metrics and Student t 95 percent confidence intervals across replications. Open Gemini conversation	Modified	Adapted metric summaries and intervals to independent replication outputs. P95 completed waiting and unserved backlog are reported with their distinct definitions.
Gemini	Candidate service construction. Prompt summary: Calculate fixed budgets, median stop travel offsets, evenly spaced bus times, and downstream candidate arrivals. Open Gemini conversation	Modified	Used the construction ideas with midpoint spacing within each hour and the fixed scenario budget. Generic linspace examples do not reproduce published GTFS timestamps.
Gemini	Matched comparisons. Prompt summary: Evaluate alternative schedules on preserved passenger events and pair matched replications. Open Gemini conversation	Modified	Used the same input streams and paired differences. Simplified AI examples without capacity constraints and randomly generated mock waits were not used as simulation results.

AI tool	Purpose, prompt summary, and link	Evaluation	Justification
Gemini	Fitness and feasibility. Prompt summary: Define a waiting-time objective and a fitness function with feasibility constraints. Open Gemini conversation	Modified	The final search minimizes mean completed waiting with integer, bound, budget, and backlog checks. The generic knapsack profit example and its soft penalty are not the project objective.
ChatGPT	Simulation explanation. Prompt summary: Separate data preparation from the repeated event-driven simulation loop and preserve scenarios, boundary rules, and matched replications. Open ChatGPT conversation	Modified	Used the organization advice for setup, FIFO events, capacity, period-end queues, and final backlog. The earlier forecasting workflow was superseded by service optimization.
Claude	Assumptions and state review. Prompt summary: response reviews the draft assumptions, state variables, units, and event logic. Open Claude conversation	Modified	Used the need for explicit sources, units, and executable rules. The final scope excludes random delay, dwell feedback, stochastic cancellation, and emergent bunching instead of assigning unsupported parameters.
Claude	Demand and AI evaluation limits. Prompt summary: reviews forecast inputs, demand undercount, and evaluation. Open Claude conversation	Modified	Retained the distinction between modeled unmet demand and true hidden demand. Suggested forecast metrics do not apply to the current GA role because no demand predictor is trained.
Claude	Disclosure completeness. Prompt summary: response calls for a per-prompt disclosure log and group responsibility. Open Claude conversation	Accepted	The disclosure identifies tools, purposes, prompts, links, and evaluation. The shared page hides its attachments, so it does not independently verify the original draft files.
Gemini	Reduced-service removal checks. Prompt summary: Explain validation for removing one trip per study hour using the Stop 6 midpoint rule and fixed period budgets. Open Gemini conversation	Modified	Used hourly removal, earlier-time tie, cross-stop trip_id, and 38/39-budget checks. The response reference to Stop 4 is corrected to the selected sequences 6 to 10.
Gemini	High-demand streams. Prompt summary: Explain rounding, baseline passenger preservation, additional arrivals, and seed consistency for a 40 percent increase. Open Gemini conversation	Modified	Multiply the source means by 1.40 before half-up rounding. Preserve baseline IDs and arrival times and add only extra passengers. Baseline boarding times and realized alightings need not remain unchanged after demand changes.

AI tool	Purpose, prompt summary, and link	Evaluation	Justification
Gemini	Combined-scenario loads. Prompt summary: Assess whether reduced-service occupancy and alighting treatment can be reused without multiplying those targets by 1.40. Open Gemini conversation	Modified	The increase applies to selected-stop boardings. Rebuild per-bus assignments under reduced service while retaining hourly upstream and assigned alighting targets. Onboard capping can change realized alightings.
Gemini	Candidate validity. Prompt summary: Explain feasibility checks for every five-hour GA trip allocation. Open Gemini conversation	Modified	Check five finite integers, hourly bounds, exact budgets, and the backlog safeguard. Mathematical count feasibility does not prove actual vehicle, crew, or dispatch feasibility.
Gemini	Search and selection. Prompt summary: Assess using a small fixed search replication set followed by larger-set evaluation. Open Gemini conversation	Modified	A small matched search set reduces cost. Larger checks used to choose a plan remain selection data; they are not unbiased out-of-sample ground truth. The final project separately evaluates frozen plans on unused seeds.
Gemini	Reduced-service hourly bounds. Prompt summary: Critically compare 5 to 12, 4 to 11, and 4 to 12 without choosing based on better GA results. Open Gemini conversation	Modified	Selected 4 to 12 for reduced service within the fixed 38/39 budgets. The lower bound includes the reduced reference counts; the upper bound retains the modeled 12-trip option.
Gemini	Reference and operational fairness. Prompt summary: Assess whether the reference belongs in the feasible space and whether 4 to 12 can introduce bias. Open Gemini conversation	Modified	The reference count vector fits the bounds. Midpoint-spaced candidates still differ from its original timestamps. A GTFS maximum does not prove fleet feasibility; concentrated service can create local waiting tradeoffs.
Claude	GA scenario consistency. Prompt summary: Check generalized GA scenario inputs, reviewer revisions, explanation quality, and consistency with reference notebooks. Open Claude conversation	Modified	Used advice on scenario-specific inputs, caches, unchanged hourly flow targets, and reproducible engine checks. The shared review describes an earlier population, search budget, bounds, and result set, not the final settings.
Claude	Paired scenario evaluation. Prompt summary: Check generalized GA scenario inputs, reviewer revisions, explanation quality, and consistency with reference notebooks. Open Claude conversation	Modified	Used paired replication comparisons with 95 percent intervals and backlog reporting. Small differences in average backlog alone do not establish statistical non-inferiority.

AI tool	Purpose, prompt summary, and link	Evaluation	Justification
Claude	Regularized reference control. Prompt summary: Check generalized GA scenario inputs, reviewer revisions, explanation quality, and consistency with reference notebooks. Open Claude conversation	Accepted	The evenly spaced reference-count control separates some timing effects from hourly allocation effects. The supplied conversation does not document advice to add the later demand-rule control.
Claude	Search noise and validity. Prompt summary: Check generalized GA scenario inputs, reviewer revisions, explanation quality, and consistency with reference notebooks. Open Claude conversation	Modified	Reported optimizer seed variation and added validity checks. The final 100-plan GA search remains limited; three seeds and a larger budget do not establish a global optimum.
Claude	Comparison tables and wording. Prompt summary: Provide a comparison table, then summarize the comparison. Open Claude conversation	Modified	Used the comparison structure and advice on wording and citations. Earlier numerical tables are not copied into final results, and Claude explicitly checked cited works against notebook lists rather than publishers.

11.4 AI Responsibility Statement

All group members remain responsible for verifying code correctness, validating assumptions, interpreting results, evaluating AI-generated suggestions, and defending all aspects of the project. Using AI tools does not transfer responsibility for submitted work to the AI system.

Table 22. Member signatures

Member	Signature	Date
Elijah Crisehea M. Nollen		10/05/2026
Ma. Sofia Anne C. Revilla		10/05/2026
Karol Joy M. Ronquillo		10/05/2026

12. Conclusion

This simulation project connects recorded passenger activity and scheduled service to reproducible queue outcomes. Frozen GA allocations reduce mean completed wait against the regularized reference in all eight condition-period comparisons. Results against the demand rule are mixed, so GA is not consistently the best simple alternative. The timing decomposition shows why published-reference gains cannot be attributed entirely to hourly allocation. Final backlog describes cutoff coverage in the current tests; some local waits and period-end queues worsen despite improved period means. The practical contribution is a transparent comparison of service candidates and their tradeoffs. The limits in Section 2.2 and further tests in Section 9 bound its use as conditional planning evidence.

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Appendix A. Detailed Independent Test Results

Table 23. Tail waits queues and backlog against published reference

Condition and period	Metric	Reference	GA	Difference	95 percent CI
Baseline Morning	P95 wait	9.063	7.307	-1.756	[-1.878, -1.634]
Baseline Morning	Peak queue	17.800	17.433	-0.367	[-1.465, 0.732]
Baseline Morning	Final backlog	1.633	1.200	-0.433	[-0.687, -0.180]
Baseline Evening	P95 wait	7.802	7.184	-0.618	[-0.853, -0.382]
Baseline Evening	Peak queue	9.833	8.433	-1.400	[-2.084, -0.716]
Baseline Evening	Final backlog	3.567	0.767	-2.800	[-3.543, -2.057]
High Demand Morning	P95 wait	9.035	7.986	-1.049	[-1.134, -0.965]
High Demand Morning	Peak queue	23.067	22.000	-1.067	[-2.400, 0.267]
High Demand Morning	Final backlog	2.600	1.967	-0.633	[-0.919, -0.348]
High Demand Evening	P95 wait	7.827	7.155	-0.672	[-0.865, -0.479]
High Demand Evening	Peak queue	14.000	11.500	-2.500	[-3.192, -1.808]
High Demand Evening	Final backlog	5.233	1.067	-4.167	[-4.933, -3.400]
Reduced Service Morning	P95 wait	13.920	8.336	-5.585	[-5.863, -5.306]
Reduced Service Morning	Peak queue	23.733	18.333	-5.400	[-6.789, -4.011]
Reduced Service Morning	Final backlog	1.633	1.567	-0.067	[-0.161, 0.028]
Reduced Service Evening	P95 wait	11.657	8.632	-3.025	[-3.368, -2.682]
Reduced Service Evening	Peak queue	10.500	8.833	-1.667	[-2.327, -1.006]

Condition and period	Metric	Reference	GA	Difference	95 percent CI
Reduced Service Evening	Final backlog	3.567	0.767	-2.800	[-3.543, -2.057]
Combined Morning	P95 wait	13.903	8.305	-5.599	[-5.883, -5.314]
Combined Morning	Peak queue	32.000	24.400	-7.600	[-9.269, -5.931]
Combined Morning	Final backlog	2.600	2.533	-0.067	[-0.161, 0.028]
Combined Evening	P95 wait	11.838	9.070	-2.768	[-3.119, -2.417]
Combined Evening	Peak queue	14.433	11.833	-2.600	[-3.400, -1.800]
Combined Evening	Final backlog	5.233	2.400	-2.833	[-3.477, -2.190]

Note. Source: review_holdout_paired_comparisons.csv. Every row uses 30 matched replications. Wait values are minutes; queue and backlog values are passengers. Difference is GA minus published reference. These are pointwise intervals.

Table 24. Period boundary queues and post-event occupancy

Condition	Period	Reference end queue	GA end queue	Reference occupancy	GA occupancy
Baseline	Morning	3.50	2.63	15.72	15.73
Baseline	Evening	3.57	2.63	12.21	12.26
High Demand	Morning	5.13	3.93	17.72	17.74
High Demand	Evening	5.23	2.87	13.81	13.88
Reduced Service	Morning	3.50	4.30	17.79	17.80
Reduced Service	Evening	3.57	2.63	13.78	13.83
Combined	Morning	5.13	6.20	20.05	20.05
Combined	Evening	5.23	5.93	15.58	15.63

Note. Sources: independent replication tables. All values are passengers. End queue is measured at 11:00 or 20:00; final backlog in Appendix A follows all eligible events. Occupancy is an event average.

Appendix B. Experiment Settings

Table 25. Search and evaluation settings

Setting	Current value
Modeled stops and capacity	Stop sequences 6 to 10; assumed capacity 60 passengers.
Periods	Morning 06:00 to 11:00; evening 15:00 to 20:00.
Reference hourly counts	Morning (5, 12, 12, 8, 6); evening (8, 12, 11, 8, 5).
Reduced hourly counts	Morning (4, 11, 11, 7, 5); evening (7, 11, 10, 7, 4).
GA population and parent selection	Final population 12; screen population 6; tournaments of up to 3.
GA preservation and variation	Best 2 plans preserved; per-gene parent choice probability 0.5; neighbor mutation probability 0.5.
Final GA budget	100 unique plans per condition, period, and optimizer seed.
Optimizer seeds	101, 202, 303.
Search fitness replications	First 5 matched passenger replications.
Scenario and selection precision	Start 30; add 10 to maximum 50; 95 percent Student t intervals for mean wait and peak queue.
Precision targets	Half-width at most 1 below mean 10; otherwise at most 10 percent of the mean.
Independent test	30 unused seeds; SeedSequence([2026, 90902]); replication IDs 1001 to 1030.
SA screen settings	Initial temperature 0.20; cooling rate 0.93.
PSO screen settings	Swarm size 6; inertia 0.70; cognitive and social coefficients 1.40.
Selection safeguard	Mean final backlog no greater than matched reference on search and larger selection sets.

Note. Sources: baseline and GA notebooks and technical documentation. Screen settings do not describe the final 100-plan GA budget.

Appendix C. Execution and Runtime

Environment setup, package installation, and execution instructions are provided in the repository README: <https://github.com/elijahnollen/m66-passenger-queue-simulation>.

A measured sequential run used Python 3.12.14. Table 26 reports one observed run, including notebook execution and output writes. Package installation and downloading files are excluded. Times are machine-specific estimates rather than guaranteed runtimes; interactive Jupyter execution can differ.

Table 26. Measured sequential execution times

Stage	Seconds	Minutes
m66_dataset_analysis.ipynb	2.9	0.05
m66_baseline_simulation.ipynb	41.1	0.69
m66_scenario_simulation.ipynb	8.4	0.14
m66_ga_optimization.ipynb	348.2	5.80
plot_results.py	1.2	0.02
Full sequential run	401.8	6.70

Note. Source: execution_runtime.csv. Measured UTC: 2026-10-04T15:14:54.620556+00:00. Full sequential run includes the stages above. The CSV records environment details and a fingerprint of code cells and Python modules.

Appendix D. Dataset Acquisition and Filtering Evidence

These screenshots record the actual source-acquisition steps already documented by the group. The live source counts describe that acquisition snapshot and can change as the public dataset is updated. The frozen raw CSV and local filtering counts are the reproducible inputs for the current simulation.

Table 27. Source acquisition and local filtering counts

Stage	Records
Full MTA dataset	155387648
2025 only	92202449
2025 + M66	165888
2025 + M66 + Eastbound	86959
Weekdays only	64531
Study hours only	35372
Selected stop sequences 6-10	12815
Final cleaned selected dataset	12814

Note. Source: filtering_summary.csv. The first three rows describe website acquisition. The frozen eastbound CSV starts at 86,959 records; later rows are local filters.

Figure 9. MTA source dataset overview

What's in this Dataset?

Rows: **155M** Columns: **9**

Columns (9)

Column Name	Description	API Field Name	Data Type
Date	Represents the time period in which the metrics are being calculated (formatted as yyyy-mm-dd).	date	Floating Timestamp
Hour	Represents the hour of when the metrics are being calculated (formatted as yyyy-mm-dd HH:MM AM/PM)	hour	Text
Route ID	Identifies each individual bus route, as well as cumulative totals for all bus routes (identified as ALL).	route_id	Text
Direction	Identifies the direction the bus is going on the route.	direction	Text
Stop ID	Identifies each individual bus stop along the route using a distinct numeric code.	stop_id	Number
Stop Sequence	Identifier for where in a bus route's sequence of stops the route-stop is.	stop_sequence	Number
Boardings	Amount of passengers that boarded the bus at the associated stop.	boardings	Number

Note. Original acquisition screenshot supplied in the group project source document. This records the source identity and acquisition context.

Figure 10. Unfiltered MTA source table

DATA.NY.GOV

MTA Bus Stop Level Ridership: Beginning 2024

Columns (9)

Date	Hour	Route ID	Direction	Stop ID	Stop Sequence	Boardings	Alightings	Trips
2024-09-30	2024-09-30 12:00:00 AM	B25	E	308025	2	2	0	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	307243	7	2	0	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	307506	9	4	2	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	307246	10	0	1	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	307541	14	2	0	1
2024-09-30	2024-09-30 12:00:00 AM	B25	W	302413	16	1	0	1

Records per page: 50 | 1 to 50 of 155,387,648

Note. Original acquisition screenshot. The count is historical acquisition context rather than a live count asserted on the report date.

Figure 11. Calendar year 2025 source filter

The screenshot shows the DATA.NY.GOV website with a table of transit data. The table has columns: Date, Hour, Route ID, Direction, Stop ID, and Stop Sequence. A filter is applied to the Date column, showing a date range from 2025 Jan 01 12:00:00 AM to 2025 Dec 31 11:59:59 P... The filter is labeled "is between". The table shows rows 1-100 of 92,202,449 records. The filter bar at the bottom shows "Filters | Clear all" and "Apply Success!".

Date	Hour	Route ID	Direction	Stop ID	Stop Sequence
2025-01-01	2025-01-01 12:00:00 AM	B103	E	350024	3

Note. Original acquisition screenshot. The year filter retained 92,202,449 records at acquisition.

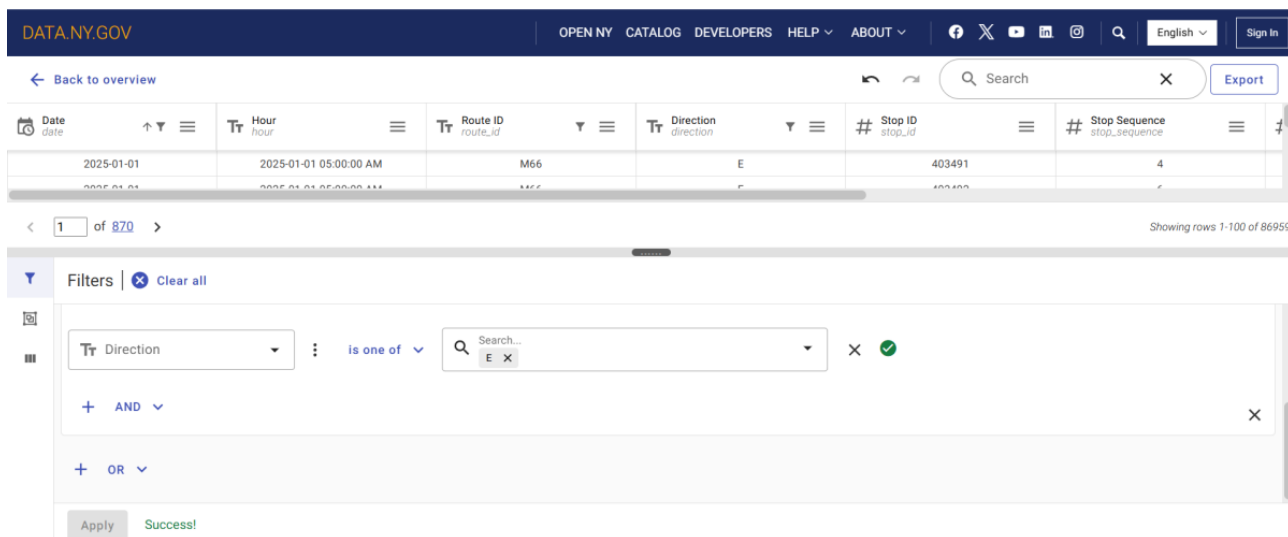
Figure 12. Year and M66 route source filters

The screenshot shows the DATA.NY.GOV website with a table of transit data. The table has columns: Date, Hour, Route ID, Direction, Stop ID, and Stop Sequence. Two filters are applied: a date range filter (2025 Jan 01 12:00:00 AM to 2025 Dec 31 11:59:59 P...) and a Route ID filter (M66). The filter is labeled "is one of". The table shows rows 1-100 of 165,888 records. The filter bar at the bottom shows "Filters | Clear all" and "Apply Success!".

Date	Hour	Route ID	Direction	Stop ID	Stop Sequence
2025-01-01	2025-01-01 05:00:00 AM	M66	E	403491	4

Note. Original acquisition screenshot. The year and route filters retained 165,888 records at acquisition.

Figure 13. Year route and eastbound source filters



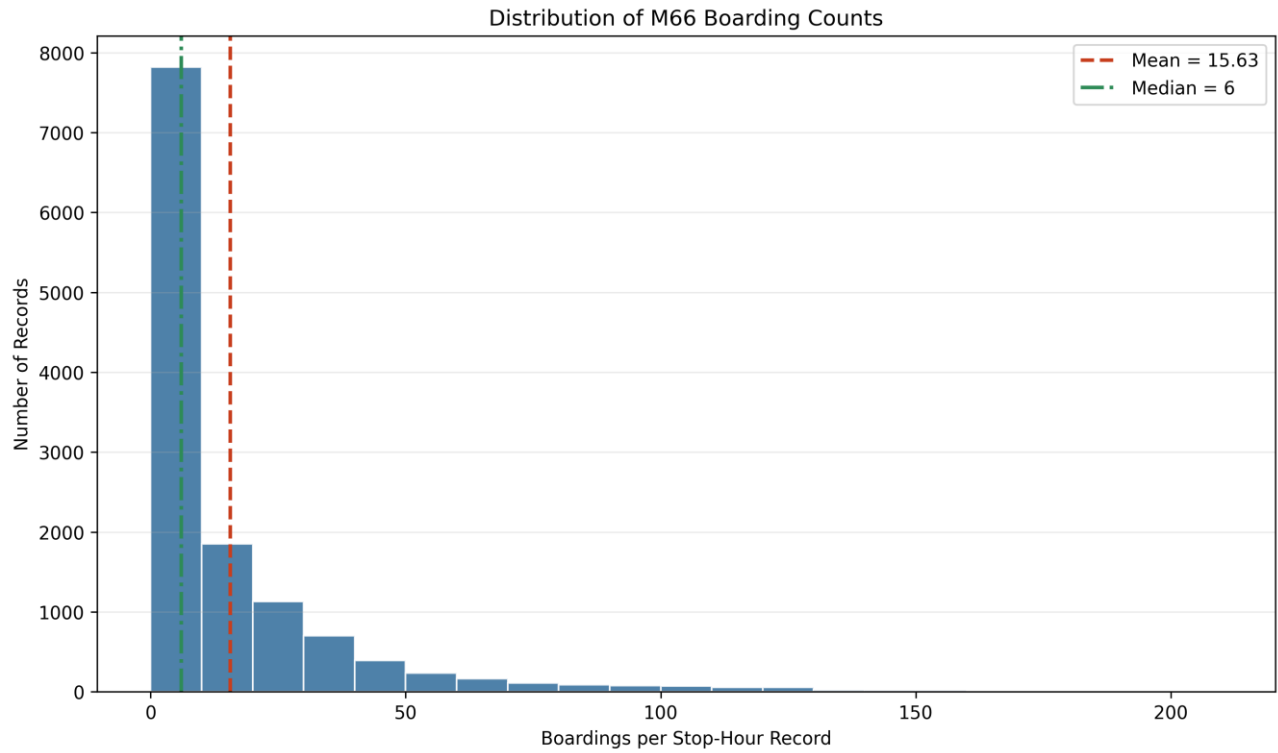
Note. Original acquisition screenshot. Eastbound filtering retained 86,959 records, matching the frozen raw extract.

Local filtering and quality-check totals are reported below. Section 3.3 explains the cleaning rules and treatment of unavailable combinations.

Table 28. Quality checks for the selected records

Check	Result
Selected records before cleaning	12815
Missing cell values	0
Exact duplicate rows	0
Logical duplicate rows	0
Negative values	0
Passenger activity with zero trips	1
Source-absent date-stop-hour combinations	235
Unavailable combinations after cleaning	236
Final cleaned records	12814

Figure 14. Distribution of recorded boarding counts



Note. Source: repository output/figures/05_boarding_count_distribution.png. This supports inspection of the right-skewed distribution. Large valid boarding counts are retained rather than deleted solely for being extreme.

Appendix E. GitHub Contribution Evidence

The commit links below provide evidence of each member's integration, scenario, and GA work summarized in Table 19. Each entry identifies the member and the work shown in the linked changes. These selected commits document specific contributions; they are not a complete authorship history.

Elijah Crisehea M. Nollen: integration of scenario and GA implementation.

<https://github.com/elijahnollen/m66-passenger-queue-simulation/commit/7fb5985bde24dd8b95f6391467288abfc7d59053?diff=split>

Ma. Sofia Anne C. Revilla: integration of scenario contexts into GA. <https://github.com/elijahnollen/m66-passenger-queue-simulation/commit/a8f7de0cab029621ec5e4fb0b39a7749f4af999b>

Ma. Sofia Anne C. Revilla: GA quality revisions and paired evaluation evidence.

<https://github.com/elijahnollen/m66-passenger-queue-simulation/commit/89ade5ad2d31c9af2fbbc0bdde41a7a6cee79617>

Karol Joy M. Ronquillo: scenario notebook rewrite. <https://github.com/elijahnollen/m66-passenger-queue-simulation/commit/5625be8baea94a558dbb97244e8c204a16c81bd9?diff=split>

Karol Joy M. Ronquillo: revised scenario figure output. <https://github.com/elijahnollen/m66-passenger-queue-simulation/commit/415b2b5f31ed6f56985b29b50a565a794b7e3893?diff=split>

Appendix F. Last Bus Diagnostics

Each listed time is the final eligible bus arrival at the specified stop. Times can extend beyond the official period end because an eligible trip completes its downstream stop events. The time vectors follow Stops 6, 7, 8, 9, and 10 in that order. Passenger counts below are sums of stop means over 30 matched independent replications. Full diagnostics include the regularized reference and demand rule as well as the published reference and GA. Compared with published service, GA has a later final bus in all eight Stop 6 cases. Across the five stops and eight cases, 32 final arrivals are later and eight are earlier. The earlier arrivals occur at Stops 7 to 10 in reduced-service and combined morning, about 9 to 12.5 seconds earlier. Baseline evening Stop 6 is 6 minutes 6 seconds later. These shifts explain why a backlog comparison cannot be read as a capacity-only effect.

Table 29. Last eligible bus times and final backlog components

Condition and period	Method	Last arrivals at Stops 6 to 10	Late arrivals	Earlier unserved	Final backlog
Baseline Morning	Ref.	10:53:48, 11:00:51, 11:02:00, 11:03:38, 11:08:00	1.633	0.000	1.633
Baseline Morning	GA	10:55:00, 11:01:42, 11:02:48, 11:04:26, 11:08:48	1.200	0.000	1.200
Baseline Evening	Ref.	19:48:54, 19:52:25, 19:53:00, 19:54:06, 19:57:00	3.567	0.000	3.567
Baseline Evening	GA	19:55:00, 19:59:56, 20:00:44, 20:01:50, 20:04:44	0.767	0.000	0.767
High Demand Morning	Ref.	10:53:48, 11:00:51, 11:02:00, 11:03:38, 11:08:00	2.600	0.000	2.600
High Demand Morning	GA	10:55:00, 11:01:42, 11:02:48, 11:04:26, 11:08:48	1.967	0.000	1.967
High Demand Evening	Ref.	19:48:54, 19:52:25, 19:53:00, 19:54:06, 19:57:00	5.233	0.000	5.233
High Demand Evening	GA	19:55:43, 20:00:39, 20:01:27, 20:02:33, 20:05:27	1.067	0.000	1.067
Reduced Service Morning	Ref.	10:53:48, 11:00:51, 11:02:00, 11:03:38, 11:08:00	1.633	0.000	1.633
Reduced Service Morning	GA	10:54:00, 11:00:42, 11:01:48, 11:03:26, 11:07:48	1.567	0.000	1.567
Reduced Service Evening	Ref.	19:48:54, 19:52:25, 19:53:00, 19:54:06, 19:57:00	3.567	0.000	3.567
Reduced Service Evening	GA	19:55:00, 19:59:56, 20:00:44, 20:01:50, 20:04:44	0.767	0.000	0.767

Condition and period	Method	Last arrivals at Stops 6 to 10	Late arrivals	Earlier unserved	Final backlog
Combined Morning	Ref.	10:53:48, 11:00:51, 11:02:00, 11:03:38, 11:08:00	2.600	0.000	2.600
Combined Morning	GA	10:54:00, 11:00:42, 11:01:48, 11:03:26, 11:07:48	2.533	0.000	2.533
Combined Evening	Ref.	19:48:54, 19:52:25, 19:53:00, 19:54:06, 19:57:00	5.233	0.000	5.233
Combined Evening	GA	19:52:30, 19:57:26, 19:58:14, 19:59:20, 20:02:14	2.400	0.000	2.400

Note. Source: review_last_bus_summary.csv. Ref. means published reference. Times are rounded to the nearest second for display; CSV times retain minute precision. Counts are passengers. Late arrivals include passengers arriving at the exact bus timestamp under bus-first event priority. Earlier unserved is the final backlog component from arrivals strictly before the last bus, not the number ever denied boarding.